

Link.Foundation.Links.Notations Class Library

1.1 ./Link.Foundation.Links.Notations/Comments.cs

```
1 namespace Link.Foundation.Links.Notations
2 {
3     /// <summary>
4     /// Comments, and the rule that decides where one starts.
5     /// </summary>
6     /// <remarks>
7     /// <c>#</c> starts a comment when it opens a token: at the start of the document or
8     /// after a space, a tab or a line break. A <c>#</c> written inside a token
9     /// (<c>issue#1047</c>) or inside a delimited reference (<c>"# not a comment"</c>) is
10    /// an ordinary character.
11    /// </remarks>
12    public static class Comments
13    {
14        /// <summary>
15        /// The character that starts a comment.
16        /// </summary>
17        public const char Comment = '#';
18
19        /// <summary>
20        /// What a delimited reference may start after.
21        /// </summary>
22        private const string BeforeReference = " \t\n\r(";
23
24        /// <summary>
25        /// What a comment may start after.
26        /// </summary>
27        private const string BeforeComment = " \t\n\r";
28
29        /// <summary>
30        /// Blanks every comment of a document.
31        /// </summary>
32        /// <param name="document">The document to read.</param>
33        /// <returns>
34        /// The same document with the characters of every comment replaced by spaces.
35        /// </returns>
36        /// <remarks>
37        /// Every character of a comment is replaced rather than removed, so every later
38        /// character keeps the position it had in the document the caller wrote, and
39        /// anything the parser reports about a position still points at what the reader
40        /// can see.
41        /// </remarks>
42        public static string StripComments(string document)
43        {
44            char[]? blanked = null;
45            int position = 0;
46
47            while (position < document.Length)
48            {
49                char character = document[position];
50
51                if (Quotes.IsQuote(character) && Follows(document, position, BeforeReference))
52                {
53                    int end = Quotes.ReferenceEnd(document, position);
54                    position = end < 0 ? position + 1 : end;
55                    continue;
56                }
57
58                if (character == Comment && Follows(document, position, BeforeComment))
59                {
60                    blanked ??= document.ToCharArray();
61                    while (position < document.Length &&
62                        document[position] != '\n' &&
63                        document[position] != '\r')
64                    {
65                        blanked[position] = ' ';
66                        position++;
67                    }
68                    continue;
69                }
70
71                position++;
72            }
73
74            return blanked == null ? document : new string(blanked);
75        }
76    }
77 }
```

```

78     /// <summary>
79     /// Reports whether the character before the position is one of the allowed ones,
80     /// the start of the document counting as allowed.
81     /// </summary>
82     private static bool Follows(string document, int position, string allowed) =>
83         position == 0 || allowed.IndexOf(document[position - 1]) >= 0;
84 }
85 }

```

1.2 ./Link.Foundation.Links.Notation/FormatConfig.cs

```

1 namespace Link.Foundation.Links.Notation
2 {
3     /// <summary>
4     /// FormatConfig is an alias for FormatOptions to maintain API consistency
5     /// with the Python implementation.
6     ///
7     /// C# uses 'FormatOptions' internally, but provides 'FormatConfig'
8     /// as an alias to match Python's naming convention.
9     /// </summary>
10    public class FormatConfig : FormatOptions
11    {
12        // FormatConfig inherits all functionality from FormatOptions
13        // No additional implementation needed - this is purely for naming consistency
14    }
15 }

```

1.3 ./Link.Foundation.Links.Notation/FormatOptions.cs

```

1 using System;
2
3 namespace Link.Foundation.Links.Notation
4 {
5     /// <summary>
6     /// FormatOptions for Lino notation formatting.
7     /// Provides configuration options for controlling how Link objects are formatted.
8     /// </summary>
9     public class FormatOptions
10    {
11        /// <summary>
12        /// If true, omit parentheses where safe (default: false)
13        /// </summary>
14        public bool LessParentheses { get; set; } = false;
15
16        /// <summary>
17        /// Maximum line length before auto-indenting (default: 80)
18        /// </summary>
19        public int MaxLineLength { get; set; } = 80;
20
21        /// <summary>
22        /// If true, indent lines exceeding MaxLineLength (default: false)
23        /// </summary>
24        public bool IndentLongLines { get; set; } = false;
25
26        /// <summary>
27        /// Maximum number of references before auto-indenting (default: null = unlimited)
28        /// </summary>
29        public int? MaxInlineRefs { get; set; } = null;
30
31        /// <summary>
32        /// If true, group consecutive links with same ID (default: false)
33        /// </summary>
34        public bool GroupConsecutive { get; set; } = false;
35
36        /// <summary>
37        /// String to use for indentation (default: "  " = two spaces)
38        /// </summary>
39        public string IndentString { get; set; } = "  ";
40
41        /// <summary>
42        /// If true, prefer inline format when under thresholds (default: true)
43        /// </summary>
44        public bool PreferInline { get; set; } = true;
45
46        /// <summary>
47        /// Check if line should be indented based on length
48        /// </summary>
49        /// <param name="line">The line to check</param>
50        /// <returns>True if line should be indented based on length threshold</returns>
51        public bool ShouldIndentByLength(string line)

```

```

52     {
53         if (!IndentLongLines)
54         {
55             return false;
56         }
57         // Count printable unicode characters
58         return line.Length > MaxLineLength;
59     }
60
61     /// <summary>
62     /// Check if link should be indented based on reference count
63     /// </summary>
64     /// <param name="refCount">Number of references in the link</param>
65     /// <returns>True if link should be indented based on reference count threshold</returns>
66     public bool ShouldIndentByRefCount(int refCount)
67     {
68         if (MaxInlineRefs == null)
69         {
70             return false;
71         }
72         return refCount > MaxInlineRefs.Value;
73     }
74 }
75 }

```

1.4 ./Link.Foundation.Links.Notation/FurthestFailureTracer.cs

```

1  using Pegasus.Common;
2  using Pegasus.Common.Tracing;
3
4  namespace Link.Foundation.Links.Notation
5  {
6      /// <summary>
7      /// Remembers the furthest position a parse reached.
8      /// </summary>
9      /// <remarks>
10     /// A parsing expression grammar backtracks, so the cursor the generated parser raises
11     /// its error with is the start of the document rather than the place the document
12     /// stopped making sense. The furthest position any rule reached is that place, and the
13     /// tracer is the only hook the generated parser offers to observe it.
14     /// </remarks>
15     internal sealed class FurthestFailureTracer : ITracer
16     {
17         /// <summary>Offset of the furthest position the parse reached.</summary>
18         public int Furthest { get; private set; }
19
20         public void TraceRuleEnter(string ruleName, Cursor cursor) => Reach(cursor.Location);
21
22         public void TraceRuleExit<T>(string ruleName, Cursor cursor, IParseResult<T> result)
23         {
24             if (result != null) Reach(result.EndCursor.Location);
25         }
26
27         public void TraceCacheHit<T>(string ruleName, Cursor cursor, CacheKey key,
↪ IParseResult<T> result)
28         {
29             if (result != null) Reach(result.EndCursor.Location);
30         }
31
32         public void TraceCacheMiss(string ruleName, Cursor cursor, CacheKey key)
33         {
34         }
35
36         public void TraceInfo(string ruleName, Cursor cursor, string info)
37         {
38         }
39
40         private void Reach(int location)
41         {
42             if (location > Furthest) Furthest = location;
43         }
44     }
45 }

```

1.5 ./Link.Foundation.Links.Notation/ILinksGroupListExtensions.cs

```

1  using System.Collections.Generic;
2  using System.Runtime.CompilerServices;
3
4  namespace Link.Foundation.Links.Notation
5  {

```

```

6     /// <summary>
7     /// Provides extension methods for collections of <see cref="LinksGroup{TLinkAddress}"/>
↪ instances.
8     /// </summary>
9     public static class ILinksGroupListExtensions
10    {
11        /// <summary>
12        /// Converts a collection of links groups to a flat list of links by processing each
↪ group hierarchically.
13        /// </summary>
14        /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This
↪ can be any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
15        /// <param name="groups">The collection of links groups to convert.</param>
16        /// <returns>A flat list of links representing all the links from the input
↪ groups.</returns>
17        [MethodImpl(MethodImplOptions.AggressiveInlining)]
18        public static List<Link<TLinkAddress>> ToLinksList<TLinkAddress>(this
↪ IList<LinksGroup<TLinkAddress>> groups)
19        {
20            var list = new List<Link<TLinkAddress>>();
21            for (var i = 0; i < groups.Count; i++)
22            {
23                CollectLinksWithIndentedIdSyntaxSupport(list, groups[i], null);
24            }
25            return list;
26        }
27
28        private static void
↪ CollectLinksWithIndentedIdSyntaxSupport<TLinkAddress>(List<Link<TLinkAddress>> list,
↪ LinksGroup<TLinkAddress> group, Link<TLinkAddress>? parentDependency)
29        {
30            var link = group.Link;
31            var groups = group.Groups;
32
33            if (groups != null && groups.Count > 0)
34            {
35                bool isIndentedIdSyntax = link.Id != null &&
36                                         (link.Values == null || link.Values.Count == 0);
37
38                if (isIndentedIdSyntax && !parentDependency.HasValue)
39                {
40                    var childValues = new List<Link<TLinkAddress>>();
41                    for (int i = 0; i < groups.Count; i++)
42                    {
43                        var transformedLink = TransformIndentedIdLink(groups[i]);
44                        childValues.Add(transformedLink);
45                    }
46
47                    var linkWithChildren = new Link<TLinkAddress>(link.Id, childValues);
48                    list.Add(linkWithChildren);
49                }
50                else
51                {
52                    var transformedGroups = new List<LinksGroup<TLinkAddress>>();
53                    for (int i = 0; i < groups.Count; i++)
54                    {
55                        var childGroup = groups[i];
56                        var childLink = childGroup.Link;
57                        var childGroups = childGroup.Groups;
58
59                        if (childLink.Id != null &&
60                            (childLink.Values == null || childLink.Values.Count == 0) &&
61                            childGroups != null && childGroups.Count > 0)
62                        {
63                            var transformedLink = TransformIndentedIdLink(childGroup);
64                            transformedGroups.Add(new LinksGroup<TLinkAddress>(transformedLink));
65                        }
66                        else
67                        {
68                            transformedGroups.Add(childGroup);
69                        }
70                    }
71
72                    var currentDependency = parentDependency.HasValue ?
↪ parentDependency.Value.Combine(link) : link;
73                    list.Add(currentDependency);
74
75                    for (int i = 0; i < transformedGroups.Count; i++)
76                    {

```

```

77         CollectLinksWithIndentedIdSyntaxSupport(list, transformedGroups[i],
↪     currentDependency);
78     }
79 }
80 }
81 else
82 {
83     var currentLink = parentDependency.HasValue ?
↪     parentDependency.Value.Combine(link) : link;
84     list.Add(currentLink);
85 }
86 }
87
88 private static Link<TLinkAddress>
↪ TransformIndentedIdLink<TLinkAddress>(LinksGroup<TLinkAddress> group)
89 {
90     var link = group.Link;
91     var groups = group.Groups;
92
93     if (groups != null && groups.Count > 0 &&
94         link.Id != null &&
95         (link.Values == null || link.Values.Count == 0))
96     {
97         var childValues = new List<Link<TLinkAddress>>();
98         for (int i = 0; i < groups.Count; i++)
99         {
100             var transformedChild = TransformIndentedIdLink(groups[i]);
101             childValues.Add(transformedChild);
102         }
103         return new Link<TLinkAddress>(link.Id, childValues);
104     }
105     else if (groups != null && groups.Count > 0)
106     {
107         var values = new List<Link<TLinkAddress>>();
108         if (link.Values != null)
109         {
110             values.AddRange(link.Values);
111         }
112         for (int i = 0; i < groups.Count; i++)
113         {
114             values.Add(TransformIndentedIdLink(groups[i]));
115         }
116         return new Link<TLinkAddress>(link.Id, values);
117     }
118     else if (!group.IsParenthesized && link.Id == null && link.Values != null &&
↪     link.Values.Count == 1)
119     {
120         return link.Values[0];
121     }
122     else
123     {
124         return link;
125     }
126 }
127 }
128 }

```

1.6 ./Link.Foundation.Links.Notation/IListExtensions.cs

```

1 using Platform.Collections;
2 using System;
3 using System.Collections.Generic;
4 using System.Linq;
5 using System.Runtime.CompilerServices;
6
7 namespace Link.Foundation.Links.Notation
8 {
9     /// <summary>
10     /// Provides extension methods for formatting collections of <see
↪     cref="Link{TLinkAddress}" /> instances.
11     /// </summary>
12     public static class IListExtensions
13     {
14         /// <summary>
15         /// Formats a collection of links as a multi-line string with each link on a separate
↪     line.
16         /// </summary>
17         /// <typeparam name="TLinkAddress">The type used for link
↪     addresses/identifiers.</typeparam>
18         /// <param name="links">The collection of links to format.</param>

```

```

19     /// <returns>A multi-line string representation of the links.</returns>
20     [MethodImpl(MethodImplOptions.AggressiveInlining)]
21     public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links) =>
↪     string.Join(Environment.NewLine, links.Select(l => l.ToString()));
22
23     /// <summary>
24     /// Formats a collection of links as a multi-line string with optional parentheses
↪     trimming for cleaner output.
25     /// </summary>
26     /// <typeparam name="TLinkAddress">The type used for link
↪     addresses/identifiers.</typeparam>
27     /// <param name="links">The collection of links to format.</param>
28     /// <param name="lessParentheses">True to remove outer parentheses from each link for
↪     cleaner formatting; false to use standard formatting.</param>
29     /// <returns>A multi-line string representation of the links.</returns>
30     [MethodImpl(MethodImplOptions.AggressiveInlining)]
31     public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links, bool
↪     lessParentheses)
32     {
33         if (lessParentheses == false)
34         {
35             return links.Format();
36         }
37         else
38         {
39             ↪     return string.Join(Environment.NewLine, links.Select(l =>
↪     l.ToString().TrimSingle('(').TrimSingle(')')));
40         }
41     }
42
43     /// <summary>
44     /// Formats a collection of links using FormatOptions configuration.
45     /// </summary>
46     /// <typeparam name="TLinkAddress">The type used for link
↪     addresses/identifiers.</typeparam>
47     /// <param name="links">The collection of links to format.</param>
48     /// <param name="options">The FormatOptions to use for formatting.</param>
49     /// <returns>A multi-line string representation of the links.</returns>
50     [MethodImpl(MethodImplOptions.AggressiveInlining)]
51     public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links,
↪     FormatOptions options)
52     {
53         if (links == null || links.Count == 0)
54         {
55             return string.Empty;
56         }
57
58         // Apply consecutive link grouping if enabled
59         var linksToFormat = options.GroupConsecutive ? GroupConsecutiveLinks(links) : links;
60
61         // Format each link with options
62         var formattedLinks = linksToFormat.Select(l => l.FormatWithOptions(options));
63         return string.Join(Environment.NewLine, formattedLinks);
64     }
65
66     /// <summary>
67     /// Group consecutive links with the same ID.
68     /// </summary>
69     private static List<Link<TLinkAddress>>
↪     GroupConsecutiveLinks<TLinkAddress>(IList<Link<TLinkAddress>> links)
70     {
71         if (links == null || links.Count == 0)
72         {
73             return new List<Link<TLinkAddress>>();
74         }
75
76         var grouped = new List<Link<TLinkAddress>>();
77         int i = 0;
78
79         while (i < links.Count)
80         {
81             var current = links[i];
82
83             // Look ahead for consecutive links with same ID
84             if (current.Id != null && current.Values != null && current.Values.Count > 0)
85             {
86                 // Collect all values with same ID
87                 var sameIdValues = new List<Link<TLinkAddress>>(current.Values);

```

```

88         int j = i + 1;
89
90         while (j < links.Count)
91         {
92             var nextLink = links[j];
93             if (EqualityComparer<TLinkAddress>.Default.Equals(nextLink.Id,
↪ current.Id) &&
94                 nextLink.Values != null && nextLink.Values.Count > 0)
95             {
96                 sameIdValues.AddRange(nextLink.Values);
97                 j++;
98             }
99             else
100             {
101                 break;
102             }
103         }
104
105         // If we found consecutive links, create grouped link
106         if (j > i + 1)
107         {
108             var groupedLink = new Link<TLinkAddress>(current.Id, sameIdValues);
109             grouped.Add(groupedLink);
110             i = j;
111             continue;
112         }
113     }
114
115     grouped.Add(current);
116     i++;
117 }
118
119     return grouped;
120 }
121 }
122 }

```

1.7 ./Link.Foundation.Links.Notation/Link.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Runtime.CompilerServices;
4  using System.Text;
5  using Platform.Collections;
6  using Platform.Collections.Lists;
7
8  namespace Link.Foundation.Links.Notation
9  {
10     /// <summary>
11     /// Represents a link in the Links Notation with an optional identifier and nested values.
12     /// Links can represent simple references, complex nested structures, or relationships
↪ between references to links.
13     /// </summary>
14     /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
↪ any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
15     public struct Link<TLinkAddress>: IEquatable<Link<TLinkAddress>>
16     {
17         private static readonly EqualityComparer<TLinkAddress> EqualityComparerInstance =
↪ EqualityComparer<TLinkAddress>.Default;
18
19         /// <summary>
20         /// Gets the identifier or address of this link. Can be null for anonymous links.
21         /// </summary>
22         public readonly TLinkAddress? Id;
23
24         /// <summary>
25         /// Gets the collection of nested link values. Can be null or empty for simple reference
↪ links.
26         /// </summary>
27         public readonly IList<Link<TLinkAddress>>? Values;
28
29         /// <summary>
30         /// Initializes a new link with the specified identifier and values.
31         /// </summary>
32         /// <param name="id">The link identifier or address.</param>
33         /// <param name="values">The nested link values.</param>
34         [MethodImpl(MethodImplOptions.AggressiveInlining)]
35         public Link(TLinkAddress? id, IList<Link<TLinkAddress>>? values) => (Id, Values) = (id,
↪ values);
36
37         /// <summary>

```

```

38     /// Initializes a new anonymous link with the specified values.
39     /// </summary>
40     /// <param name="values">The nested link values.</param>
41     [MethodImpl(MethodImplOptions.AggressiveInlining)]
42     public Link(ICollection<Link<TLinkAddress>> values) : this(default!, values) { }
43
44     /// <summary>
45     /// Initializes a new anonymous link with the specified array of values.
46     /// </summary>
47     /// <param name="values">The nested link values as a parameter array.</param>
48     [MethodImpl(MethodImplOptions.AggressiveInlining)]
49     public Link(params Link<TLinkAddress>[] values) : this(default!, values) { }
50
51     /// <summary>
52     /// Initializes a new simple reference link with the specified identifier.
53     /// </summary>
54     /// <param name="id">The link identifier or address.</param>
55     [MethodImpl(MethodImplOptions.AggressiveInlining)]
56     public Link(TLinkAddress id) : this(id, default!) { }
57
58     /// <summary>
59     /// Returns the string representation of this link in Links Notation format.
60     /// </summary>
61     /// <returns>A string representation of the link with proper escaping and
62     ↪ formatting.</returns>
63     [MethodImpl(MethodImplOptions.AggressiveInlining)]
64     public override string ToString() => Id == null ?
65         $"({GetValuesString()})" :
66         (Values == null || Values.Count == 0) ?
67             $"({EscapeReference(Id.ToString())})" :
68             $"({EscapeReference(Id.ToString())}: {GetValuesString()})";
69
70     /// <summary>
71     /// Gets the string representation of the link's values.
72     /// </summary>
73     /// <returns>A space-separated string of the values, or empty string if no values
74     ↪ exist.</returns>
75     [MethodImpl(MethodImplOptions.AggressiveInlining)]
76     public string GetValuesString()
77     {
78         if (Values == null || Values.Count == 0)
79         {
80             return "";
81         }
82         var sb = new StringBuilder();
83         for (int i = 0; i < Values.Count; i++)
84         {
85             if (i > 0)
86             {
87                 sb.Append(' ');
88             }
89             sb.Append(GetValueString(Values[i]));
90         }
91         return sb.ToString();
92     }
93
94     /// <summary>
95     /// Simplifies the link structure by unwrapping single-value containers and recursively
96     ↪ simplifying nested values.
97     /// </summary>
98     /// <returns>A simplified version of this link.</returns>
99     [MethodImpl(MethodImplOptions.AggressiveInlining)]
100     public Link<TLinkAddress> Simplify()
101     {
102         if (Values == null || Values.Count == 0)
103         {
104             return this;
105         }
106         else if (Values.Count == 1)
107         {
108             return Values[0];
109         }
110         else
111         {
112             var newValues = new Link<TLinkAddress>[Values.Count];
113             for (int i = 0; i < Values.Count; i++)
114             {
115                 newValues[i] = Values[i].Simplify();
116             }
117             return new Link<TLinkAddress>(newValues);
118         }
119     }

```

```

113     }
114     return new Link<TLinkAddress>(Id, newValues);
115 }
116 }
117
118 /// <summary>
119 /// Combines this link with another link to create a new link containing both as values.
120 /// </summary>
121 /// <param name="other">The other link to combine with.</param>
122 /// <returns>A new link containing both links as values.</returns>
123 [MethodImpl(MethodImplOptions.AggressiveInlining)]
124 public Link<TLinkAddress> Combine(Link<TLinkAddress> other) => new
↪ Link<TLinkAddress>(this, other);
125
126 /// <summary>
127 /// Gets the string representation of a link value, choosing the appropriate format.
128 /// </summary>
129 /// <param name="value">The link value to convert to string.</param>
130 /// <returns>The string representation of the link value.</returns>
131 [MethodImpl(MethodImplOptions.AggressiveInlining)]
132 public static string GetValueString(Link<TLinkAddress> value) =>
↪ value.ToLinkOrIdString();
133
134 /// <summary>
135 ↪ /// Escapes a reference string for safe use in Links Notation format by adding quotes if
136 ↪ necessary.
137 /// </summary>
138 /// <param name="reference">The reference string to escape.</param>
139 /// <returns>The escaped reference string with appropriate quoting.</returns>
140 public static string EscapeReference(string? reference)
141 {
142     if (reference == null)
143     {
144         return "";
145     }
146     // The empty reference is written as a bare delimiter pair, so that it reads
147     // back as itself instead of disappearing from the document.
148     if (reference.Length == 0)
149     {
150         return "\"\"";
151     }
152     // A reference that begins with a "#" has to be quoted, or it would read
153     // back as a comment. A "#" anywhere else in a reference is content
154     // ("issue#1047"), so only the first character matters.
155     if (
156         reference.StartsWith("#") ||
157         reference.Contains(":") ||
158         reference.Contains("(") ||
159         reference.Contains(")") ||
160         reference.Contains(" ") ||
161         reference.Contains("\t") ||
162         reference.Contains("\n") ||
163         reference.Contains("\r") ||
164         reference.Contains("\\"))
165     {
166         return $"{reference}";
167     }
168     else if (reference.Contains("\""))
169     {
170         return $"\"{reference}\"";
171     }
172     else
173     {
174         return reference;
175     }
176 }
177
178 /// <summary>
179 ↪ /// Converts the link to its string representation, choosing between simple reference
180 ↪ format or full link format.
181 /// </summary>
182 /// <returns>A string representation optimized for the link's content.</returns>
183 [MethodImpl(MethodImplOptions.AggressiveInlining)]
184 public string ToLinkOrIdString() => (Values == null || Values.Count == 0) ? (Id == null
↪ ? "" : EscapeReference(Id?.ToString())) : ToString();
185
186 /// <summary>

```

```

186     /// Implicitly converts a value to a simple reference link.
187     /// </summary>
188     /// <param name="value">The value to convert.</param>
189     [MethodImpl(MethodImplOptions.AggressiveInlining)]
190     public static implicit operator Link<TLinkAddress>(TLinkAddress value) => new
↪ Link<TLinkAddress>(value);

191     /// <summary>
192     /// Implicitly converts a tuple of identifier and values to a link.
193     /// </summary>
194     /// <param name="value">The tuple containing identifier and values.</param>
195     [MethodImpl(MethodImplOptions.AggressiveInlining)]
196     public static implicit operator Link<TLinkAddress>((TLinkAddress,
↪ IList<Link<TLinkAddress>>) value) => new(value.Item1, value.Item2);

198     /// <summary>
199     /// Implicitly converts a source-target pair to an anonymous link.
200     /// </summary>
201     /// <param name="value">The tuple containing source and target links.</param>
202     [MethodImpl(MethodImplOptions.AggressiveInlining)]
203     public static implicit operator Link<TLinkAddress>((Link<TLinkAddress> source,
↪ Link<TLinkAddress> target) value) => new(value.source, value.target);

205     /// <summary>
206     /// Implicitly converts a tuple of identifier, source, and target to a link.
207     /// </summary>
208     /// <param name="value">The tuple containing id, source and target.</param>
209     [MethodImpl(MethodImplOptions.AggressiveInlining)]
210     public static implicit operator Link<TLinkAddress>((id<TLinkAddress> id,
↪ Link<TLinkAddress> source, Link<TLinkAddress> target) value) => new(value.id.Id, new[] {
↪ value.source, value.target });

212     /// <summary>
213     /// Implicitly converts a source-linker-target triplet to an anonymous link.
214     /// </summary>
215     /// <param name="value">The tuple containing source, linker, and target links.</param>
216     [MethodImpl(MethodImplOptions.AggressiveInlining)]
217     public static implicit operator Link<TLinkAddress>((Link<TLinkAddress> source,
↪ Link<TLinkAddress> linker, Link<TLinkAddress> target) value) => new(value.source,
↪ value.linker, value.target);

219     /// <summary>
220     /// Implicitly converts a tuple of identifier, source, linker, and target to a link.
221     /// </summary>
222     /// <param name="value">The tuple containing id, source, linker, and target.</param>
223     [MethodImpl(MethodImplOptions.AggressiveInlining)]
224     public static implicit operator Link<TLinkAddress>((id<TLinkAddress> id,
↪ Link<TLinkAddress> source, Link<TLinkAddress> linker, Link<TLinkAddress> target) value) =>
↪ new(value.id.Id, new[] { value.source, value.linker, value.target });

226     /// <summary>
227     /// Determines whether the specified object is equal to this link.
228     /// </summary>
229     /// <param name="obj">The object to compare with this link.</param>
230     /// <returns>True if the specified object is equal to this link; otherwise,
↪ false.</returns>
232     [MethodImpl(MethodImplOptions.AggressiveInlining)]
233     public override bool Equals(object? obj) => obj is Link<TLinkAddress> link &&
↪ Equals(link);

234     /// <summary>
235     /// Returns the hash code for this link.
236     /// </summary>
237     /// <returns>A hash code for this link.</returns>
238     [MethodImpl(MethodImplOptions.AggressiveInlining)]
239     public override int GetHashCode() => (Id, (Values ??
↪ Array.Empty<Link<TLinkAddress>>()).GenerateHashCode()).GetHashCode();

241     /// <summary>
242     /// Indicates whether the current link is equal to another link.
243     /// Two anonymous links (both with null Ids) are considered equal if their values are
↪ equal.
245     /// </summary>
246     /// <param name="other">The link to compare with this link.</param>
247     /// <returns>True if the current link is equal to the other parameter; otherwise,
↪ false.</returns>
248     [MethodImpl(MethodImplOptions.AggressiveInlining)]

```

```

249     public bool Equals(Link<TLinkAddress> other)
250     {
251         // Both have null IDs - compare only values
252         if (Id == null && other.Id == null)
253         {
254             return (Values ?? Array.Empty<Link<TLinkAddress>>()).EqualTo(other.Values ??
↪ Array.Empty<Link<TLinkAddress>>());
255         }
256
257         // Only one has null ID - not equal
258         if (Id == null || other.Id == null)
259         {
260             return false;
261         }
262
263         // Both have IDs - compare IDs and values
264         return EqualityComparerInstance.Equals(Id, other.Id) &&
265             (Values ?? Array.Empty<Link<TLinkAddress>>()).EqualTo(other.Values ??
↪ Array.Empty<Link<TLinkAddress>>());
266     }
267
268     /// <summary>
269     /// Determines whether two link instances are equal.
270     /// </summary>
271     /// <param name="left">The first link to compare.</param>
272     /// <param name="right">The second link to compare.</param>
273     /// <returns>True if the links are equal; otherwise, false.</returns>
274     [MethodImpl(MethodImplOptions.AggressiveInlining)]
275     public static bool operator ==(Link<TLinkAddress> left, Link<TLinkAddress> right) =>
↪ left.Equals(right);
276
277     /// <summary>
278     /// Determines whether two link instances are not equal.
279     /// </summary>
280     /// <param name="left">The first link to compare.</param>
281     /// <param name="right">The second link to compare.</param>
282     /// <returns>True if the links are not equal; otherwise, false.</returns>
283     [MethodImpl(MethodImplOptions.AggressiveInlining)]
284     public static bool operator !=(Link<TLinkAddress> left, Link<TLinkAddress> right) =>
↪ !(left == right);
285 }
286 }

```

1.8 ./Link.Foundation.Links.Notation/LinkFormatExtensions.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Runtime.CompilerServices;
5  using System.Text;
6
7  namespace Link.Foundation.Links.Notation
8  {
9      /// <summary>
10     /// Provides extension methods for formatting <see cref="Link{TLinkAddress}"/> instances
↪ with FormatOptions.
11     /// </summary>
12     public static class LinkFormatExtensions
13     {
14         /// <summary>
15         /// Format a link using FormatOptions configuration.
16         /// </summary>
17         /// <typeparam name="TLinkAddress">The type used for link
↪ addresses/identifiers.</typeparam>
18         /// <param name="link">The link to format.</param>
19         /// <param name="options">The FormatOptions to use.</param>
20         /// <returns>Formatted string representation.</returns>
21         public static string FormatWithOptions<TLinkAddress>(this Link<TLinkAddress> link,
↪ FormatOptions options)
22         {
23             // Empty link
24             if (link.Id == null && (link.Values == null || link.Values.Count == 0))
25             {
26                 return options.LessParentheses ? "" : "()";
27             }
28
29             // Link with only ID, no values
30             if (link.Values == null || link.Values.Count == 0)
31             {

```

```

32     var escapedId = Link<TLinkAddress>.EscapeReference(link.Id?.ToString());
33     return options.LessParentheses && !NeedsParentheses(link.Id?.ToString()) ?
34         escapedId :
35         $"({escapedId})";
36 }
37
38 // Check if we should use indented format
39 bool shouldIndent = false;
40 if (options.ShouldIndentByRefCount(link.Values.Count))
41 {
42     shouldIndent = true;
43 }
44 else
45 {
46     // Try inline format first
47     var valuesStr = string.Join(" ", link.Values.Select(v => GetValueString(v)));
48     string testLine;
49     if (link.Id != null)
50     {
51         var idStr = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
52         testLine = options.LessParentheses ? $"{idStr}: {valuesStr}" : $"{idStr}:
↪ {valuesStr}";
53     }
54     else
55     {
56         testLine = options.LessParentheses ? valuesStr : $"({valuesStr})";
57     }
58
59     if (options.ShouldIndentByLength(testLine))
60     {
61         shouldIndent = true;
62     }
63 }
64
65 // Format with indentation if needed
66 if (shouldIndent && options.PreferInline == false)
67 {
68     return FormatIndented(link, options);
69 }
70
71 // Standard inline formatting
72 var values = string.Join(" ", link.Values.Select(v => GetValueString(v)));
73
74 // Link with values only (null id)
75 if (link.Id == null)
76 {
77     if (options.LessParentheses)
78     {
79         var allSimple = link.Values.All(v => v.Values == null || v.Values.Count ==
↪ 0);
80         if (allSimple)
81         {
82             return string.Join(" ", link.Values.Select(v =>
↪ Link<TLinkAddress>.EscapeReference(v.Id?.ToString())));
83         }
84         return values;
85     }
86     return $"({values})";
87 }
88
89 // Link with ID and values
90 var id = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
91 var withColon = $"{id}: {values}";
92 return options.LessParentheses && !NeedsParentheses(link.Id?.ToString()) ?
93     withColon :
94     $"({withColon})";
95 }
96
97 /// <summary>
98 /// Format a link with indentation.
99 /// </summary>
100 private static string FormatIndented<TLinkAddress>(Link<TLinkAddress> link,
↪ FormatOptions options)
101 {
102     // FormatIndented is only reached with a non-empty Values collection,
103     // but the nullable annotation on Link<T>.Values does not know that.
104     var values = link.Values ?? Array.Empty<Link<TLinkAddress>>();
105
106     if (link.Id == null)

```

```

107     {
108         // Values only - format each on separate line
109         var lines = values.Select(v => options.IndentString + GetValueString(v));
110         return string.Join(Environment.NewLine, lines);
111     }
112
113     // Link with ID - format as id:\n value1\n value2
114     var idStr = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
115     var sb = new StringBuilder();
116     sb.Append($"{{idStr}}:");
117
118     foreach (var v in values)
119     {
120         sb.Append(Environment.NewLine);
121         sb.Append(options.IndentString);
122         sb.Append(GetValueString(v));
123     }
124
125     return sb.ToString();
126 }
127
128 /// <summary>
129 /// Get the string representation of a value.
130 /// </summary>
131 private static string GetValueString<TLinkAddress>(Link<TLinkAddress> value)
132 {
133     return value.ToLinkOrIdString();
134 }
135
136 /// <summary>
137 /// Check if a string needs to be wrapped in parentheses.
138 /// </summary>
139 private static bool NeedsParentheses(string? s)
140 {
141     return s != null && (s.Contains(" ") || s.Contains(":") || s.Contains("(") ||
↪ s.Contains(")"));
142 }
143 }
144 }

```

1.9 ./Link.Foundation.Links.Notation/LinksGroup.cs

```

1 using System;
2 using System.Collections.Generic;
3 using System.Runtime.CompilerServices;
4 using Platform.Collections;
5 using Platform.Collections.Lists;
6
7 namespace Link.Foundation.Links.Notation
8 {
9     /// <summary>
10     /// Represents a group of links with hierarchical structure, where each group contains a
↪ primary link and optional nested groups.
11     /// This structure supports the indentation-based syntax of the Links Notation.
12     /// </summary>
13     /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
↪ any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
14     public struct LinksGroup<TLinkAddress> : IEquatable<LinksGroup<TLinkAddress>>
15     {
16         /// <summary>
17         /// Gets or sets the primary link of this group.
18         /// </summary>
19         public Link<TLinkAddress> Link
20         {
21             [MethodImpl(MethodImplOptions.AggressiveInlining)]
22             get;
23             [MethodImpl(MethodImplOptions.AggressiveInlining)]
24             set;
25         }
26
27         /// <summary>
28         /// Gets or sets the collection of nested groups within this group. Can be null if no
↪ nested groups exist.
29         /// </summary>
30         public IList<LinksGroup<TLinkAddress>>? Groups
31         {
32             [MethodImpl(MethodImplOptions.AggressiveInlining)]
33             get;
34             [MethodImpl(MethodImplOptions.AggressiveInlining)]
35             set;

```

```

36     }
37
38     /// <summary>
39     /// Gets or sets a value indicating whether this group was written as a parenthesized
40     group.
41     /// A parenthesized group is a link in its own right, so it keeps its identifier and
42     values
43     /// instead of being unwrapped when it is used as a child of the indented identifier
44     syntax.
45     /// This is a parse-time hint and therefore takes no part in equality comparisons.
46     /// </summary>
47     internal bool IsParenthesized
48     {
49         [MethodImpl(MethodImplOptions.AggressiveInlining)]
50         get;
51         [MethodImpl(MethodImplOptions.AggressiveInlining)]
52         set;
53     }
54
55     /// <summary>
56     /// Initializes a new links group with the specified link and nested groups.
57     /// </summary>
58     /// <param name="link">The primary link of this group.</param>
59     /// <param name="groups">The nested groups within this group.</param>
60     [MethodImpl(MethodImplOptions.AggressiveInlining)]
61     public LinksGroup(Link<TLinkAddress> link, IList<LinksGroup<TLinkAddress>>? groups)
62     {
63         Link = link;
64         Groups = groups;
65     }
66
67     /// <summary>
68     /// Initializes a new links group with the specified link and no nested groups.
69     /// </summary>
70     /// <param name="link">The primary link of this group.</param>
71     [MethodImpl(MethodImplOptions.AggressiveInlining)]
72     public LinksGroup(Link<TLinkAddress> link) : this(link, null) { }
73
74     /// <summary>
75     /// Implicitly converts a links group to a list of links by flattening the hierarchical
76     structure.
77     /// </summary>
78     /// <param name="value">The links group to convert.</param>
79     [MethodImpl(MethodImplOptions.AggressiveInlining)]
80     public static implicit operator List<Link<TLinkAddress>>(LinksGroup<TLinkAddress> value)
81     => value.ToLinksList();
82
83     /// <summary>
84     /// Converts this links group to a flat list of links, preserving the hierarchical
85     relationships.
86     /// </summary>
87     /// <returns>A list of links representing the flattened structure of this
88     group.</returns>
89     [MethodImpl(MethodImplOptions.AggressiveInlining)]
90     public List<Link<TLinkAddress>> ToLinksList()
91     {
92         var list = new List<Link<TLinkAddress>>();
93         AppendToLinksList(list);
94         return list;
95     }
96
97     /// <summary>
98     /// Appends this links group to the specified list as flattened links.
99     /// </summary>
100    /// <param name="list">The list to append the links to.</param>
101    [MethodImpl(MethodImplOptions.AggressiveInlining)]
102    public void AppendToLinksList(List<Link<TLinkAddress>> list) => AppendToLinksList(list,
103    Link, this);
104
105    /// <summary>
106    /// Recursively appends a links group to a list, combining dependencies with nested
107    groups to create a flattened structure.
108    /// </summary>
109    /// <param name="list">The list to append the links to.</param>
110    /// <param name="dependency">The dependency link to combine with nested groups.</param>
111    /// <param name="group">The links group to process.</param>
112    [MethodImpl(MethodImplOptions.AggressiveInlining)]
113    public static void AppendToLinksList(List<Link<TLinkAddress>> list, Link<TLinkAddress>
114    dependency, LinksGroup<TLinkAddress> group)

```

```

105     {
106         list.Add(dependency);
107         var groups = group.Groups;
108         if (groups != null && !groups.IsNullOrEmpty())
109         {
110             for (int i = 0; i < groups.Count; i++)
111             {
112                 var innerGroup = groups[i];
113                 AppendToLinksList(list, dependency.Combine(innerGroup.Link), innerGroup);
114             }
115         }
116     }
117
118     /// <summary>
119     /// Determines whether the specified object is equal to this links group.
120     /// </summary>
121     /// <param name="obj">The object to compare with this links group.</param>
122     /// <returns>True if the specified object is equal to this links group; otherwise,
123     ↪ false.</returns>
124     [MethodImpl(MethodImplOptions.AggressiveInlining)]
125     public override bool Equals(object? obj) => obj is LinksGroup<TLinkAddress> linksGroup ?
126     ↪ Equals(linksGroup) : false;
127
128     /// <summary>
129     /// Returns the hash code for this links group.
130     /// </summary>
131     /// <returns>A hash code for this links group.</returns>
132     [MethodImpl(MethodImplOptions.AggressiveInlining)]
133     public override int GetHashCode() => (Link, (Groups ??
134     ↪ Array.Empty<LinksGroup<TLinkAddress>>()).GenerateHashCode()).GetHashCode();
135
136     /// <summary>
137     /// Indicates whether the current links group is equal to another links group.
138     /// </summary>
139     /// <param name="other">The links group to compare with this links group.</param>
140     /// <returns>True if the current links group is equal to the other parameter; otherwise,
141     ↪ false.</returns>
142     [MethodImpl(MethodImplOptions.AggressiveInlining)]
143     public bool Equals(LinksGroup<TLinkAddress> other) => Link.Equals(other.Link) && (Groups
144     ↪ ?? Array.Empty<LinksGroup<TLinkAddress>>()).EqualTo(other.Groups ??
145     ↪ Array.Empty<LinksGroup<TLinkAddress>>());
146
147     /// <summary>
148     /// Determines whether two links group instances are equal.
149     /// </summary>
150     /// <param name="left">The first links group to compare.</param>
151     /// <param name="right">The second links group to compare.</param>
152     /// <returns>True if the links groups are equal; otherwise, false.</returns>
153     [MethodImpl(MethodImplOptions.AggressiveInlining)]
154     public static bool operator ==(LinksGroup<TLinkAddress> left, LinksGroup<TLinkAddress>
155     ↪ right) => left.Equals(right);
156
157     /// <summary>
158     /// Determines whether two links group instances are not equal.
159     /// </summary>
160     /// <param name="left">The first links group to compare.</param>
161     /// <param name="right">The second links group to compare.</param>
162     /// <returns>True if the links groups are not equal; otherwise, false.</returns>
163     [MethodImpl(MethodImplOptions.AggressiveInlining)]
164     public static bool operator !=(LinksGroup<TLinkAddress> left, LinksGroup<TLinkAddress>
165     ↪ right) => !(left == right);
166
167     /// <summary>
168     /// Returns a string representation of this links group.
169     /// </summary>
170     /// <returns>A string representation of this links group.</returns>
171     [MethodImpl(MethodImplOptions.AggressiveInlining)]
172     public override string ToString() => ToLinksList().Format();
173 }

```

1.10 ./Link.Foundation.Links.Notation/ParseException.cs

```

1  using System;
2  using System.Text;
3
4  namespace Link.Foundation.Links.Notation
5  {
6      /// <summary>

```

```

7  /// The error raised when a document does not parse, carrying the position the parser
8  /// stopped at.
9  /// </summary>
10 /// <remarks>
11 /// The position is the furthest one the parser reached, which is the character the
12 /// document stops making sense at rather than the point the last alternative gave up
13 /// on. It derives from <see cref="FormatException"/>, the exception the generated
14 /// parser raised on its own, so code that catches that keeps working.
15 /// </remarks>
16 public class ParseException : FormatException
17 {
18     /// <summary>The number of characters a quoted line is cut down to.</summary>
19     private const int QuotedLineWidth = 80;
20
21     /// <summary>What a message writes in place of the part of a long line it left
22     out.</summary>
23     private const string Ellipsis = "...";
24
25     /// <summary>
26     /// Creates an exception describing where <paramref name="subject"/> stopped parsing.
27     /// </summary>
28     /// <param name="subject">The document that failed to parse.</param>
29     /// <param name="offset">Offset of the position the parser stopped at.</param>
30     /// <param name="innerException">The error the generated parser raised, if any.</param>
31     public ParseException(string subject, int offset, Exception? innerException = null)
32         : this(Locate(subject ?? string.Empty, offset), innerException)
33     {
34     }
35
36     private ParseException(Position position, Exception? innerException)
37         : base(Describe(position), innerException)
38     {
39         Offset = position.Offset;
40         Line = position.Line;
41         Column = position.Column;
42         Found = position.Found;
43         LineText = position.LineText;
44     }
45
46     /// <summary>Offset of the offending position from the start of the document.</summary>
47     public int Offset { get; }
48
49     /// <summary>Line the offending position is on, counted from 1.</summary>
50     public int Line { get; }
51
52     /// <summary>Column the offending position is at, counted from 1.</summary>
53     public int Column { get; }
54
55     /// <summary>The character found instead, or null at the end of the document.</summary>
56     public char? Found { get; }
57
58     /// <summary>The offending line, as written, without its line ending.</summary>
59     public string LineText { get; }
60
61     /// <summary>The one-line summary: where the parser stopped and what stands
62     there.</summary>
63     public string Summary => Summarize(new Position(Offset, Line, Column, Found, LineText));
64
65     /// <summary>
66     /// The offending line with a caret under the offending column, quoted the way a
67     /// compiler quotes source. A long line is shown as a window around the caret, so the
68     /// message stays the same size whether the document has ten lines or fifteen hundred.
69     /// </summary>
70     public string Snippet => Quote(new Position(Offset, Line, Column, Found, LineText));
71
72     /// <summary>Where the parser stopped, and what the document holds there.</summary>
73     private readonly record struct Position(
74         int Offset,
75         int Line,
76         int Column,
77         char? Found,
78         string LineText);
79
80     /// <summary>The message the exception carries: the summary and the quoted
81     line.</summary>
82     private static string Describe(Position position) =>
83         $"Syntax error at {Summarize(position)}\n{Quote(position)}";
84
85     private static string Summarize(Position position)

```

```

83 {
84     var found = position.Found.HasValue
85         ? $"{Escape(position.Found.Value)}\"
86         : "end of input";
87     return $"line {position.Line}, column {position.Column}: unexpected {found}";
88 }
89
90 private static string Quote(Position position)
91 {
92     var (quoted, column) = QuoteLine(position.LineText, position.Column);
93     var number = position.Line.ToString();
94     var gutter = new string(' ', number.Length);
95     return $"{number} | {quoted}\n{gutter} | {new string(' ', column - 1)}~";
96 }
97
98 private static string Escape(char character) => character switch
99 {
100     '\n' => "\\n",
101     '\r' => "\\r",
102     '\t' => "\\t",
103     '"' => "\"\"",
104     '\\' => "\\\\",
105     => character.ToString(),
106 };
107
108 /// <summary>
109 /// Turns the position the parser stopped at into a line, a column and the line
110 /// itself, so the message can point at the defect instead of quoting the rest of the
111 /// document.
112 /// </summary>
113 private static Position Locate(string document, int offset)
114 {
115     offset = Math.Max(0, Math.Min(offset, document.Length));
116
117     var line = 1;
118     var lineStart = 0;
119     for (var index = 0; index < offset; index++)
120     {
121         if (document[index] == '\n')
122         {
123             line++;
124             lineStart = index + 1;
125         }
126     }
127
128     var lineEnd = document.IndexOf('\n', lineStart);
129     if (lineEnd < 0) lineEnd = document.Length;
130
131     var lineText = document.Substring(lineStart, lineEnd - lineStart).TrimEnd('\r');
132     var found = offset < document.Length ? document[offset] : (char?)null;
133     return new Position(offset, line, offset - lineStart + 1, found, lineText);
134 }
135
136 /// <summary>
137 /// Cuts <paramref name="line"/> down to a window around <paramref name="column"/>, and
138 /// says which column the offending character sits at in that window. Both columns
139 /// count from 1.
140 /// </summary>
141 private static (string Quoted, int Column) QuoteLine(string line, int column)
142 {
143     if (line.Length <= QuotedLineWidth) return (line, column);
144
145     var target = column - 1;
146     var lastStart = line.Length - QuotedLineWidth;
147     var start = Math.Min(Math.Max(target - (QuotedLineWidth / 2), 0), lastStart);
148     var end = start + QuotedLineWidth;
149
150     var quoted = new StringBuilder();
151     if (start > 0) quoted.Append(Ellipsis);
152     quoted.Append(line, start, QuotedLineWidth);
153     if (end < line.Length) quoted.Append(Ellipsis);
154
155     var shift = start > 0 ? Ellipsis.Length : 0;
156     return (quoted.ToString(), target - start + shift + 1);
157 }
158 }
159 }

```

1.11 ./Link.Foundation.Links.Notation/Parser.cs

```

1  using System;
2  using System.Collections.Generic;
3
4  namespace Link.Foundation.Links.Notation
5  {
6      /// <summary>
7      /// Parses Links Notation documents into links.
8      /// </summary>
9      /// <remarks>
10     /// The rules live in <c>Parser.peg</c> and are compiled into
11     /// <see cref="GeneratedParser"/>; this class runs them and, when a document does not
12     /// parse, reports the position the parser stopped at instead of the bare
13     /// <see cref="FormatException"/> the generated parser raises.
14     /// </remarks>
15     public class Parser
16     {
17         /// <summary>
18         /// Creates a parser.
19         /// </summary>
20         /// <param name="comments">
21         /// When <see langword="false"/>, <c>#</c> is read as an ordinary character
22         /// instead of the start of a comment.
23         /// </param>
24         public Parser(bool comments = true) => Comments = comments;
25
26         /// <summary>
27         /// Whether <c>#</c> starts a comment that runs to the end of its line.
28         /// </summary>
29         public bool Comments { get; }
30
31         /// <summary>
32         /// Parses a Links Notation document.
33         /// </summary>
34         /// <param name="subject">The document to parse.</param>
35         /// <param name="fileName">The optional file name to use in error messages.</param>
36         /// <returns>The links the document holds.</returns>
37         /// <exception cref="ParseException">
38         /// Thrown when the document does not parse. The exception carries the line, the
39         /// column and the offending line of the position the parser stopped at.
40         /// </exception>
41         public IList<Link<string>> Parse(string subject, string? fileName = null)
42         {
43             // Comments are blanked rather than removed, so a position reported for the
44             // prepared document is the same position in the document the caller wrote.
45             var prepared = Comments ? Notation.Comments.StripComments(subject) : subject;
46             var tracer = new FurthestFailureTracer();
47             var parser = new GeneratedParser { Tracer = tracer };
48             try
49             {
50                 return parser.Parse(prepared, fileName);
51             }
52             catch (FormatException error) when (error is not ParseException)
53             {
54                 throw new ParseException(subject, tracer.Furthest, error);
55             }
56         }
57     }
58 }

```

1.12 ./Link.Foundation.Links.Notation/Quotes.cs

```

1  using System.Text;
2
3  namespace Link.Foundation.Links.Notation
4  {
5      /// <summary>
6      /// Reads a delimited reference, the shape shared by the grammar and by
7      /// <see cref="Comments"/>.
8      /// </summary>
9      /// <remarks>
10     /// Any number N of delimiters opens and closes a reference, and 2N delimiters stand
11     /// for N written as content. The grammar reads references through this class, so
12     /// blanking comments reaches a reference exactly as far as the parser does.
13     /// </remarks>
14     internal static class Quotes
15     {
16         /// <summary>
17         /// Reports whether the character is one of the delimiters a reference can be
18         /// written between.

```

```

19     /// </summary>
20     /// <param name="character">The character to test.</param>
21     /// <returns><see langword="true"/> when the character opens a reference.</returns>
22     internal static bool IsQuote(char character) =>
23         character == '"' || character == '\'' || character == '`';
24
25     /// <summary>
26     /// A body written between an even run of delimiters is substantive when it
27     /// holds at least one visible character and does not straddle a parenthesis.
28     /// An even run can always be read as delimiter pairs enclosing nothing, so the
29     /// n-quote reading is only taken when it carries something the pairs cannot.
30     /// </summary>
31     /// <param name="content">The body to weigh.</param>
32     /// <returns><see langword="true"/> when the body carries something a pair
33     ↪ cannot.</returns>
34     internal static bool IsSubstantiveBody(string content)
35     {
36         int depth = 0;
37         bool hasVisible = false;
38
39         foreach (var c in content)
40         {
41             if (c == '(')
42             {
43                 depth++;
44             }
45             else if (c == ')')
46             {
47                 depth--;
48                 if (depth < 0) return false;
49             }
50             if (!char.IsWhiteSpace(c)) hasVisible = true;
51         }
52
53         return hasVisible && depth == 0;
54     }
55
56     /// <summary>
57     /// Reads the reference starting at the given position.
58     /// </summary>
59     /// <param name="subject">The whole document being read.</param>
60     /// <param name="start">The position of the opening delimiter.</param>
61     /// <param name="value">The content of the reference, without its delimiters.</param>
62     /// <param name="length">How many characters the reference spans.</param>
63     /// <returns><see langword="true"/> when a reference was read at that position.</returns>
64     ↪ internal static bool TryParse(string? subject, int start, out string? value, out int
65     ↪ length)
66     {
67         value = null;
68         length = 0;
69         if (subject == null || start >= subject.Length || !IsQuote(subject[start])) return
70         ↪ false;
71
72         char quoteChar = subject[start];
73
74         // Count opening quotes
75         int quoteCount = 0;
76         int pos = start;
77         while (pos < subject.Length && subject[pos] == quoteChar)
78         {
79             quoteCount++;
80             pos++;
81         }
82
83         bool isEvenRun = quoteCount % 2 == 0;
84
85         string openClose = new string(quoteChar, quoteCount);
86         string escapeSeq = new string(quoteChar, quoteCount * 2);
87
88         var content = new StringBuilder();
89         while (pos < subject.Length)
90         {
91             // Check for escape sequence (2*N quotes)
92             if (pos + escapeSeq.Length <= subject.Length &&
93                 string.CompareOrdinal(subject, pos, escapeSeq, 0, escapeSeq.Length) == 0)
94             {
95                 content.Append(openClose); // 2*N quotes become N quotes
96                 pos += escapeSeq.Length;
97             }
98         }
99         if (isEvenRun)
100         {
101             content.Append(openClose);
102             value = content.ToString();
103             length = content.Length;
104         }
105     }

```

```

95         continue;
96     }
97
98     // Check for closing quotes (exactly N quotes, not more)
99     if (pos + quoteCount <= subject.Length &&
100         string.CompareOrdinal(subject, pos, openClose, 0, quoteCount) == 0)
101     {
102         int afterClose = pos + quoteCount;
103         if (afterClose >= subject.Length || subject[afterClose] != quoteChar)
104         {
105             // Found valid closing
106             if (isEvenRun && !IsSubstantiveBody(content.ToString()))
107             {
108                 return SetEmptyReference(isEvenRun, quoteCount, out value, out
↪ length);
109             }
110
111             value = content.ToString();
112             length = afterClose - start;
113             return true;
114         }
115     }
116
117     // Take next character
118     content.Append(subject[pos]);
119     pos++;
120 }
121
122 // No valid closing found
123 return SetEmptyReference(isEvenRun, quoteCount, out value, out length);
124 }
125
126 /// <summary>
127 /// Finds where the reference starting at the given position ends.
128 /// </summary>
129 /// <param name="subject">The whole document being read.</param>
130 /// <param name="start">The position of the opening delimiter.</param>
131 /// <returns>
132 /// The position right after the closing delimiters, or -1 when no reference
133 /// starts there.
134 /// </returns>
135 internal static int ReferenceEnd(string subject, int start) =>
136     TryParse(subject, start, out var ignored, out var length) ? start + length : -1;
137
138 /// <summary>
139 /// A run of an even number of delimiters that does not open a reference with a
140 /// substantive body is the empty reference: the shortest reading, a bare
141 /// delimiter pair enclosing nothing, wins over a longer n-quote delimiter.
142 /// </summary>
143 private static bool SetEmptyReference(bool isEvenRun, int quoteCount, out string? value,
↪ out int length)
144 {
145     if (!isEvenRun)
146     {
147         value = null;
148         length = 0;
149         return false;
150     }
151
152     value = string.Empty;
153     length = quoteCount;
154     return true;
155 }
156 }
157 }

```

1.13 ./Link.Foundation.Links.Notation/StreamParseException.cs

```

1 using System;
2
3 namespace Link.Foundation.Links.Notation
4 {
5     /// <summary>A parser error located relative to the complete input stream.</summary>
6     public sealed class StreamParseException : FormatException
7     {
8         internal StreamParseException(ParseException error, int offset, int line, int column)
9             : base($"Stream parse error at line {line}, column {column}: {error.Message}", error)
10         {
11             Offset = offset;
12             Line = line;
13             Column = column;

```

```

14     ParseError = error;
15 }
16
17 /// <summary>The canonical parser error raised for the buffered record.</summary>
18 public ParseException ParseError { get; }
19
20 /// <summary>Absolute character offset from the start of the stream.</summary>
21 public int Offset { get; }
22
23 /// <summary>One-based line number in the complete stream.</summary>
24 public int Line { get; }
25
26 /// <summary>One-based column number in the complete stream.</summary>
27 public int Column { get; }
28 }
29 }

```

1.14 ./Link.Foundation.Links.Notation/StreamParser.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Runtime.CompilerServices;
4  using System.Text;
5  using System.Threading;
6  using System.Threading.Tasks;
7
8  namespace Link.Foundation.Links.Notation
9  {
10     /// <summary>Incrementally emits complete top-level Links Notation records.</summary>
11     public sealed class StreamParser
12     {
13         private const int DefaultMaxBufferSize = 10 * 1024 * 1024;
14         private const string BeforeReference = " \t\n\r(";
15         private const string BeforeComment = " \t\n\r";
16
17         private readonly Parser parser;
18         private readonly StringBuilder buffer = new();
19         private readonly StringBuilder currentLine = new();
20         private readonly List<Link<string>> links = new();
21         private int? baseIndentation;
22         private bool lineClassified;
23         private int offset;
24         private int line;
25         private int column;
26         private int segmentOffset;
27         private int segmentLine;
28         private bool ended;
29         private int maxBufferSize = DefaultMaxBufferSize;
30
31         /// <summary>Creates a stream backed by a default canonical parser.</summary>
32         public StreamParser() : this(new Parser()) { }
33
34         /// <summary>Creates a stream backed by <paramref name="parser"/>.</summary>
35         public StreamParser(Parser parser)
36         {
37             parser = parser ?? throw new ArgumentNullException(nameof(parser));
38             Reset();
39         }
40
41         /// <summary>Raised once for every completed link.</summary>
42         public event Action<Link<string>>? LinkParsed;
43
44         /// <summary>Whether emitted links are retained for <see cref="Finish()"/> and <see
45         ↪ cref="Drain"/>.</summary>
46         public bool Collect { get; set; } = true;
47
48         /// <summary>The largest unresolved record accepted by the stream.</summary>
49         public int MaxBufferSize
50         {
51             get => maxBufferSize;
52             set => maxBufferSize = value > 0
53                 ? value
54                 : throw new ArgumentOutOfRangeException(nameof(value), "Maximum buffer size must
55                 ↪ be positive.");
56         }
57
58         /// <summary>The point immediately after the last character written.</summary>
59         public StreamPosition Position => new(
60             offset,
61             line,
62             column,
63             buffer.Length + currentLine.Length);
64     }
65 }

```

```

62
63 /// <summary>Consumes a chunk and returns links made complete by it.</summary>
64 public IReadOnlyList<Link<string>> Write(string chunk)
65 {
66     ArgumentNullException.ThrowIfNull(chunk);
67     if (ended) throw new InvalidOperationException("Cannot write after Finish().");
68
69     var emitted = new List<Link<string>>();
70     foreach (var character in chunk)
71     {
72         currentLine.Append(character);
73         offset++;
74
75         if (character == '\n')
76         {
77             buffer.Append( currentLine);
78             currentLine.Clear();
79             lineClassified = false;
80             line++;
81             column = 1;
82         }
83         else
84         {
85             if (! lineClassified && character != ' ' && character != '\t' && character
↪ != '\r')
86             {
87                 lineClassified = true;
88                 if (!( parser.Comments && character == Comments.Comment))
89                 {
90                     StartContentLine(LeadingSpaces( currentLine), emitted);
91                 }
92             }
93             column++;
94         }
95
96         if ( buffer.Length + currentLine.Length > MaxBufferSize)
97         {
98             throw new ArgumentException(
99                 $"Buffered record exceeds maximum size of {MaxBufferSize} characters.",
100                 nameof(chunk));
101         }
102     }
103     return emitted;
104 }
105
106 /// <summary>Finishes the stream and returns all undrained links.</summary>
107 public IReadOnlyList<Link<string>> Finish() => Finish(string.Empty);
108
109 /// <summary>Consumes a final chunk and finishes the stream.</summary>
110 public IReadOnlyList<Link<string>> Finish(string chunk)
111 {
112     if (ended)
113     {
114         if (chunk.Length > 0) throw new InvalidOperationException("Cannot write after
↪ Finish().");
115         return Collect ? links.ToArray() : Array.Empty<Link<string>>();
116     }
117
118     var emitted = chunk.Length == 0
119         ? new List<Link<string>>()
120         : new List<Link<string>>(Write(chunk));
121     var document = buffer.ToString() + currentLine;
122     if (document.Length > 0)
123     {
124         IList<Link<string>> parsed;
125         try
126         {
127             parsed = parser.Parse(document);
128         }
129         catch (ParseException error)
130         {
131             throw new StreamParseException(
132                 error,
133                 segmentOffset + error.Offset,
134                 segmentLine + error.Line - 1,
135                 error.Column);
136         }
137         Publish(parsed, emitted);
138         AdvanceSegment(document);

```

```

139     }
140
141     buffer.Clear();
142     currentLine.Clear();
143     baseIndentation = null;
144     ended = true;
145     return Collect ? links.ToArray() : emitted;
146 }
147
148 /// <summary>Returns and forgets links retained since the previous call.</summary>
149 public IReadOnlyList<Link<string>> Drain()
150 {
151     var links = links.ToArray();
152     links.Clear();
153     return links;
154 }
155
156 /// <summary>Reuses the parser while preserving its options and event handlers.</summary>
157 public StreamParser Reset()
158 {
159     buffer.Clear();
160     currentLine.Clear();
161     baseIndentation = null;
162     lineClassified = false;
163     links.Clear();
164     offset = 0;
165     line = 1;
166     column = 1;
167     segmentOffset = 0;
168     segmentLine = 1;
169     ended = false;
170     return this;
171 }
172
173 /// <summary>Lazily parses a sequence of chunks as a native enumerable.</summary>
174 public static IEnumerable<Link<string>> ParseChunks(
175     IEnumerable<string> chunks,
176     Parser? parser = null)
177 {
178     var stream = new StreamParser(parser ?? new Parser()) { Collect = false };
179     foreach (var chunk in chunks)
180     {
181         foreach (var link in stream.Write(chunk)) yield return link;
182     }
183     foreach (var link in stream.Finish()) yield return link;
184 }
185
186 /// <summary>Lazily parses an asynchronous sequence of chunks.</summary>
187 public static async IAsyncEnumerable<Link<string>> ParseChunksAsync(
188     IAsyncEnumerable<string> chunks,
189     Parser? parser = null,
190     [EnumeratorCancellation] CancellationToken cancellationToken = default)
191 {
192     var stream = new StreamParser(parser ?? new Parser()) { Collect = false };
193     await foreach (var chunk in chunks.WithCancellation(cancellationToken))
194     {
195         foreach (var link in stream.Write(chunk)) yield return link;
196     }
197     foreach (var link in stream.Finish()) yield return link;
198 }
199
200 private void StartContentLine(int indentation, List<Link<string>> emitted)
201 {
202     if ( buffer.Length > 0 &&
203         baseIndentation.HasValue &&
204         indentation <= baseIndentation.Value &&
205         StructurallyComplete( buffer.ToString(), parser.Comments))
206     {
207         try
208         {
209             var document = buffer.ToString();
210             var parsed = parser.Parse(document);
211             Publish(parsed, emitted);
212             AdvanceSegment(document);
213             buffer.Clear();
214             baseIndentation = null;
215         }
216         catch (ParseException)
217         {
218             // A rejected prefix can become valid when more input arrives.

```

```

219     }
220 }
221
222     baseIndentation ??= indentation;
223 }
224
225 private void Publish(IEnumerable<Link<string>> links, List<Link<string>> emitted)
226 {
227     foreach (var link in links)
228     {
229         emitted.Add(link);
230         if (Collect) links.Add(link);
231         LinkParsed?.Invoke(link);
232     }
233 }
234
235 private void AdvanceSegment(string document)
236 {
237     segmentOffset += document.Length;
238     foreach (var character in document)
239     {
240         if (character == '\n') segmentLine++;
241     }
242 }
243
244 private static int LeadingSpaces(StringBuilder line)
245 {
246     var indentation = 0;
247     while (indentation < line.Length && line[indentation] == ' ') indentation++;
248     return indentation;
249 }
250
251 private static bool StructurallyComplete(string document, bool comments)
252 {
253     var depth = 0;
254     for (var position = 0; position < document.Length; position++)
255     {
256         var character = document[position];
257         if (Quotes.IsQuote(character) && Follows(document, position, BeforeReference))
258         {
259             var end = Quotes.ReferenceEnd(document, position);
260             if (end < 0) return false;
261             position = end - 1;
262             continue;
263         }
264         if (comments && character == Comments.Comment && Follows(document, position,
265 ↪ BeforeComment))
266         {
267             var newline = document.IndexOf('\n', position);
268             if (newline < 0) break;
269             position = newline;
270             continue;
271         }
272         if (character == '(') depth++;
273         else if (character == ')') depth--;
274     }
275     return depth == 0;
276 }
277
278 private static bool Follows(string document, int position, string allowed) =>
279     position == 0 || allowed.IndexOf(document[position - 1]) >= 0;
280
281 /// <summary>An absolute stream position and the unresolved record size.</summary>
282 public readonly record struct StreamPosition(int Offset, int Line, int Column, int Buffered);
283 }

```

1.15 ./Link.Foundation.Links.Notation/ .cs

```

1 using System.Runtime.CompilerServices;
2
3 namespace Link.Foundation.Links.Notation
4 {
5     /// <summary>
6     /// A utility struct that provides convenient implicit conversions for creating links from
7     ↪ various tuple combinations.
8     /// The underscore name follows functional programming conventions for placeholder/utility
9     ↪ types.
10    /// </summary>
11    /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
12    ↪ any typethat uniquely identifies a link, such as string, int, or Guid.</typeparam>

```

```

10 public struct <TLinkAddress>
11 {
12     /// <summary>
13     /// Gets the underlying link represented by this utility struct.
14     /// </summary>
15     public readonly Link<TLinkAddress> Link;
16
17     /// <summary>
18     /// Initializes a new instance with the specified link.
19     /// </summary>
20     /// <param name="id">The link to wrap.</param>
21     [MethodImpl(MethodImplOptions.AggressiveInlining)]
22     public (Link<TLinkAddress> id) => Link = id;
23
24     /// <summary>
25     /// Implicitly converts a link to this utility struct.
26     /// </summary>
27     /// <param name="value">The link to convert.</param>
28     [MethodImpl(MethodImplOptions.AggressiveInlining)]
29     public static implicit operator <TLinkAddress>(Link<TLinkAddress> value) => new(value);
30
31     /// <summary>
32     /// Implicitly converts a link address to this utility struct.
33     /// </summary>
34     /// <param name="id">The link address to convert.</param>
35     [MethodImpl(MethodImplOptions.AggressiveInlining)]
36     public static implicit operator <TLinkAddress>(TLinkAddress id) => new(id);
37
38     /// <summary>
39     /// Implicitly converts a tuple of address and link to this utility struct.
40     /// </summary>
41     /// <param name="value">The tuple containing address and link.</param>
42     [MethodImpl(MethodImplOptions.AggressiveInlining)]
43     public static implicit operator <TLinkAddress>((TLinkAddress, Link<TLinkAddress>)
44     ↪ value) => new Link<TLinkAddress>(value.Item1, new[] { value.Item2 });
45
46     /// <summary>
47     /// Implicitly converts a tuple of address and two links to this utility struct.
48     /// </summary>
49     /// <param name="value">The tuple containing address and two links.</param>
50     [MethodImpl(MethodImplOptions.AggressiveInlining)]
51     public static implicit operator <TLinkAddress>((TLinkAddress, Link<TLinkAddress>,
52     ↪ Link<TLinkAddress>) value) => new Link<TLinkAddress>(value.Item1, new[] { value.Item2,
53     ↪ value.Item3 });
54
55     /// <summary>
56     /// Implicitly converts a tuple of address and three links to this utility struct.
57     /// </summary>
58     /// <param name="value">The tuple containing address and three links.</param>
59     [MethodImpl(MethodImplOptions.AggressiveInlining)]
60     public static implicit operator <TLinkAddress>((TLinkAddress, Link<TLinkAddress>,
61     ↪ Link<TLinkAddress>, Link<TLinkAddress>) value) => new Link<TLinkAddress>(value.Item1, new[]
62     ↪ { value.Item2, value.Item3, value.Item4 });
63
64     /// <summary>
65     /// Implicitly converts this utility struct back to a link.
66     /// </summary>
67     /// <param name="value">The utility struct to convert.</param>
68     [MethodImpl(MethodImplOptions.AggressiveInlining)]
69     public static implicit operator Link<TLinkAddress>(<TLinkAddress> value) => value.Link;
70 }

```

1.16 ./Link.Foundation.Links.Notation/id.cs

```

1 using System.Runtime.CompilerServices;
2
3 namespace Link.Foundation.Links.Notation
4 {
5     /// <summary>
6     /// A readonly struct that explicitly represents a link identifier/address.
7     /// This type is used to distinguish between regular values and explicit identifiers in link
8     ↪ construction.
9     /// </summary>
10    /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
11    ↪ any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
12    public readonly struct id<TLinkAddress>
13    {
14        /// <summary>

```

```

13     /// Gets the identifier value.
14     /// </summary>
15     public readonly TLinkAddress Id;
16
17     /// <summary>
18     /// Initializes a new identifier with the specified value.
19     /// </summary>
20     /// <param name="id">The identifier value.</param>
21     [MethodImpl(MethodImplOptions.AggressiveInlining)]
22     public id(TLinkAddress id) => Id = id;
23
24     /// <summary>
25     /// Explicitly converts a value to an identifier struct.
26     /// </summary>
27     /// <param name="id">The value to convert to an identifier.</param>
28     [MethodImpl(MethodImplOptions.AggressiveInlining)]
29     public static explicit operator id<TLinkAddress>(TLinkAddress id) => new(id);
30 }
31 }

```

1.17 ./Link.Foundation.Links.Notation.Tests/ApiTests.cs

```

1  using System;
2  using System.Collections.Generic;
3  using Xunit;
4
5  namespace Link.Foundation.Links.Notation.Tests
6  {
7      public static class ApiTests
8      {
9          [Fact]
10         public static void IsRefTest()
11         {
12             // C# doesn't have separate Ref/Link types, but we can test simple link behavior
13             var simpleLink = new Link<string>("some value", null);
14             Assert.Equal("some value", simpleLink.Id);
15             Assert.Null(simpleLink.Values);
16         }
17
18         [Fact]
19         public static void IsLinkTest()
20         {
21             // Test link with values
22             var values = new List<Link<string>> { new Link<string>("child", null) };
23             var link = new Link<string>("id", values);
24             Assert.Equal("id", link.Id);
25             Assert.Single(link.Values!);
26             Assert.Equal("child", link.Values?[0].Id);
27         }
28
29         [Fact]
30         public static void IsRefEquivalentTest()
31         {
32             // C# doesn't have separate Ref/Link types, but we can test simple link behavior
33             var simpleLink = new Link<string>("some value", null);
34             Assert.Equal("some value", simpleLink.Id);
35             Assert.Null(simpleLink.Values);
36         }
37
38         [Fact]
39         public static void IsLinkEquivalentTest()
40         {
41             // Test link with values
42             var values = new List<Link<string>> { new Link<string>("child", null) };
43             var link = new Link<string>("id", values);
44             Assert.Equal("id", link.Id);
45             Assert.Single(link.Values!);
46             Assert.Equal("child", link.Values?[0].Id);
47         }
48
49         [Fact]
50         public static void EmptyLinkTest()
51         {
52             var link = new Link<string>(null, new List<Link<string>>());
53             var output = link.ToString();
54             Assert.Equal("()", output);
55         }
56
57         [Fact]
58         public static void SimpleLinkTest()

```

```

59 {
60     var input = "(1: 1 1)";
61     var parser = new Parser();
62     var parsed = parser.Parse(input);
63
64     // Validate regular formatting
65     var output = parsed.Format();
66     Assert.Contains("1:", output);
67     Assert.Contains("1", output);
68 }
69
70 [Fact]
71 public static void LinkWithSourceTargetTest()
72 {
73     var input = "(index: source target)";
74     var parser = new Parser();
75     var parsed = parser.Parse(input);
76
77     // Validate regular formatting
78     var output = parsed.Format();
79     Assert.Equal(input, output);
80 }
81
82 [Fact]
83 public static void LinkWithSourceTypeTargetTest()
84 {
85     var input = "(index: source type target)";
86     var parser = new Parser();
87     var parsed = parser.Parse(input);
88
89     // Validate regular formatting
90     var output = parsed.Format();
91     Assert.Equal(input, output);
92 }
93
94 [Fact]
95 public static void SingleLineFormatTest()
96 {
97     var input = "id: value1 value2";
98     var parser = new Parser();
99     var parsed = parser.Parse(input);
100
101     // The parser should handle single-line format
102     var output = parsed.Format();
103     Assert.Contains("id", output);
104     Assert.Contains("value1", output);
105     Assert.Contains("value2", output);
106 }
107
108 [Fact]
109 public static void QuotedReferencesTest()
110 {
111     var input = @"("quoted id": "value with spaces")";
112     var parser = new Parser();
113     var parsed = parser.Parse(input);
114
115     var output = parsed.Format();
116     Assert.Contains("quoted id", output);
117     Assert.Contains("value with spaces", output);
118 }
119
120 [Fact]
121 public static void QuotedReferencesParsingTest()
122 {
123     // Test that quoted references are parsed correctly
124     var input = @"("quoted id": "value with spaces")";
125     var parser = new Parser();
126     var parsed = parser.Parse(input);
127
128     // Verify parsing worked correctly
129     var output = parsed.Format();
130     Assert.Contains("quoted id", output);
131     Assert.Contains("value with spaces", output);
132 }
133
134 [Fact]
135 public static void IndentedIdSyntaxParsingTest()
136 {
137     // Test that indented ID syntax is parsed correctly

```

```

138     var indented = "id:\n  value1\n  value2";
139     var inline = "(id: value1 value2)";
140
141     var parser = new Parser();
142     var indentedParsed = parser.Parse(indented);
143     var inlineParsed = parser.Parse(inline);
144
145     // Both should produce equivalent structures
146     var indentedOutput = indentedParsed.Format();
147     var inlineOutput = inlineParsed.Format();
148     Assert.Equal(indentedOutput, inlineOutput);
149     Assert.Equal("(id: value1 value2)", indentedOutput);
150 }
151
152 [Fact]
153 public static void IndentedIdSyntaxRoundtripTest()
154 {
155     var input = "id:\n  value1\n  value2";
156     var parser = new Parser();
157     var parsed = parser.Parse(input);
158
159     // Validate that we can format with indented syntax using FormatOptions
160     var options = new FormatOptions
161     {
162         MaxInlineRefs = 1, // Force indentation with more than 1 ref
163         PreferInline = false
164     };
165     var output = parsed.Format(options);
166     Assert.Equal(input, output);
167 }
168
169 [Fact]
170 public static void MultipleIndentedIdSyntaxParsingTest()
171 {
172     // Test that multiple indented ID links are parsed correctly
173     var indented = "id1:\n  a\n  b\nid2:\n  c\n  d";
174     var inline = "(id1: a b)\n(id2: c d)";
175
176     var parser = new Parser();
177     var indentedParsed = parser.Parse(indented);
178     var inlineParsed = parser.Parse(inline);
179
180     // Both should produce equivalent structures
181     var indentedOutput = indentedParsed.Format();
182     var inlineOutput = inlineParsed.Format();
183     Assert.Equal(indentedOutput, inlineOutput);
184     Assert.Equal("(id1: a b)\n(id2: c d)", indentedOutput);
185 }
186
187 [Fact]
188 public static void MultipleIndentedIdSyntaxRoundtripTest()
189 {
190     var input = "id1:\n  a\n  b\nid2:\n  c\n  d";
191     var parser = new Parser();
192     var parsed = parser.Parse(input);
193
194     // Validate that we can format with indented syntax using FormatOptions
195     var options = new FormatOptions
196     {
197         MaxInlineRefs = 1, // Force indentation with more than 1 ref
198         PreferInline = false
199     };
200     var output = parsed.Format(options);
201     Assert.Equal(input, output);
202 }
203 }
204 }

```

1.18 ./Link.Foundation.Links.Notation.Tests/CommentsTests.cs

```

1  using System.Linq;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      /// <summary>
7      /// Conformance tests for line comments.
8      ///
9      /// https://github.com/link-foundation/links-notation/issues/301
10     ///

```

```

11  /// <c>#</c> starts a comment that runs to the end of the line, unless it sits
12  /// inside a token or inside a delimited reference. Parsers read comments by
13  /// default and can be told to treat <c>#</c> as an ordinary character again.
14  /// The table below is shared with the Rust, JavaScript, Python, Go, Java and
15  /// PHP suites, so a document written by one implementation reads the same in
16  /// all of them.
17  /// </summary>
18  public static class CommentsTests
19  {
20      /// <summary>
21      /// Renders a parsed node unambiguously: every reference is wrapped in
22      /// angle brackets.
23      /// </summary>
24      private static string Render(Link<string> node)
25      {
26          if (node.Values == null || node.Values.Count == 0)
27          {
28              return "<" + (node.Id ?? "") + ">";
29          }
30          var head = node.Id == null ? "" : "<" + node.Id + "> ";
31          return "(" + head + string.Join(" ", node.Values.Select(Render)) + ")";
32      }
33
34      private static string RenderedWith(Parser parser, string source) =>
35          string.Join("\n", parser.Parse(source).Select(Render));
36
37      private static string Rendered(string source) => RenderedWith(new Parser(), source);
38
39      private static void AssertParsesAs(string source, string expected)
40      {
41          Assert.Equal(expected, Rendered(source));
42      }
43
44      [Fact]
45      public static void ALineThatStartsWithAHashIsAComment()
46      {
47          AssertParsesAs("# a b\n", "");
48      }
49
50      [Fact]
51      public static void ACommentMayHoldAColon()
52      {
53          // The document from #301: prose with a colon used to be rejected.
54          AssertParsesAs("# a: b\n", "");
55      }
56
57      [Fact]
58      public static void ACommentMayHoldAnythingAtAll()
59      {
60          AssertParsesAs("# ) : ( \" ' ` #\n", "");
61      }
62
63      [Fact]
64      public static void ACommentEndsAtTheEndOfItsLine()
65      {
66          AssertParsesAs("# note\na: b\n", "<a>: <b>");
67      }
68
69      [Fact]
70      public static void ACommentMayFollowALink()
71      {
72          AssertParsesAs("a: b # why\n", "<a>: <b>");
73      }
74
75      [Fact]
76      public static void ACommentMayFollowAGroup()
77      {
78          AssertParsesAs("(a b) # why\n", "<a> <b>");
79      }
80
81      [Fact]
82      public static void ACommentNeedsNoClosingNewline()
83      {
84          AssertParsesAs("a: b # why", "<a>: <b>");
85      }
86
87      [Fact]
88      public static void ACommentLineInsideAnIndentedBlockIsSkipped()
89      {

```

```

90     Assert.Equal(
91         Rendered("parent\n  child\n"),
92         Rendered("parent\n  # what the child is for\n  child\n"));
93 }
94
95 [Fact]
96 public static void ACommentLineInsideAGroupIsSkipped()
97 {
98     Assert.Equal(
99         Rendered("(\\n  a\\n  b\\n)\\n"),
100         Rendered("(\\n  a\\n  # why\\n  b\\n)\\n"));
101 }
102
103 [Fact]
104 public static void ALineOfSpacesSeparatesLinksTheWayAnEmptyLineDoes()
105 {
106     // Blanking a comment leaves a line of spaces behind, so such a line
107     // has to read the way an empty line does.
108     Assert.Equal(Rendered("a\\n\\nb\\n"), Rendered("a\\n  \\nb\\n"));
109 }
110
111 [Fact]
112 public static void ADocumentOfCommentsAloneHoldsNoLinks()
113 {
114     AssertParsesAs("# one\\n# two\\n", "");
115 }
116
117 [Fact]
118 public static void AHashInsideATokenIsAnOrdinaryCharacter()
119 {
120     AssertParsesAs("issue#1047\\n", "<issue#1047>");
121 }
122
123 [Fact]
124 public static void AHashThatOpensATokenIsAnOrdinaryCharacter()
125 {
126     AssertParsesAs("a: b#c\\n", "<a>: <b#c>");
127 }
128
129 [Fact]
130 public static void AHashInsideADelimitedReferenceIsContent()
131 {
132     AssertParsesAs("<\"# not a comment\" a\\n", "<# not a comment> <a>");
133 }
134
135 [Fact]
136 public static void ACommentMayFollowADelimitedReference()
137 {
138     AssertParsesAs("<\"a\" # why\\n", "<a>");
139 }
140
141 [Fact]
142 public static void AHashInsideAMultiLineDelimitedReferenceIsContent()
143 {
144     AssertParsesAs("<\"a # b\\nc\" d\\n", "<a # b\\nc> <d>");
145 }
146
147 [Fact]
148 public static void CommentsAreOnByDefault()
149 {
150     Assert.True(new Parser().Comments);
151     AssertParsesAs("# a b\\n", "");
152 }
153
154 [Fact]
155 public static void AParserWithoutCommentsKeepsTheHash()
156 {
157     Assert.Equal("<#> <a> <b>", RenderedWith(new Parser(comments: false), "# a b\\n"));
158 }
159
160 [Fact]
161 public static void BlankingACommentKeepsTheLengthOfTheDocument()
162 {
163     Assert.Equal("a: b      \\n", Comments.StripComments("a: b # why\\n"));
164     Assert.Equal("<\"# kept\"\\n", Comments.StripComments("<\"# kept\"\\n"));
165     Assert.Equal("issue#1047\\n", Comments.StripComments("issue#1047\\n"));
166     Assert.Equal("      \\n", Comments.StripComments("# why\\n"));
167 }
168

```

```

169 [Fact]
170 public static void ACommentDoesNotMoveThePositionALaterErrorIsReportedAt()
171 {
172     var error = Assert.Throws<ParseException>(
173         () => new Parser().Parse("# a comment\nstage: rust: nextest\n"));
174
175     Assert.Equal(2, error.Line);
176     Assert.Equal(12, error.Column);
177 }
178
179 [Fact]
180 public static void AReferenceThatBeginsWithAHashIsWrittenQuoted()
181 {
182     // Without the quotes the document would read as `a` followed by a comment.
183     var document = new Link<string>(new Link<string>("a"), new
↪ Link<string>("#tag")).ToString();
184
185     Assert.Equal("(a '#tag')", document);
186     Assert.Equal("<a> <#tag>", Rendered(document));
187 }
188
189 [Fact]
190 public static void AHashThatCannotOpenACommentIsLeftUnquoted()
191 {
192     Assert.Equal("issue#1047", Link<string>.EscapeReference("issue#1047"));
193     Assert.Equal("#", Link<string>.EscapeReference("#"));
194     Assert.Equal("#ff0000", Link<string>.EscapeReference("#ff0000"));
195 }
196 }
197 }

```

1.19 ./Link.Foundation.Links.Notation.Tests/EdgeCaseParserTests.cs

```

1 using System;
2 using Xunit;
3
4 namespace Link.Foundation.Links.Notation.Tests
5 {
6     public static class EdgeCaseParserTests
7     {
8         [Fact]
9         public static void EmptyLinkTest()
10         {
11             var source = @"";
12             var parser = new Parser();
13             // Standalone ':' is now forbidden and should throw an exception
14             Assert.Throws<ParseException>(() => parser.Parse(source));
15         }
16
17         [Fact]
18         public static void EmptyLinkWithParenthesesTest()
19         {
20             var source = @"()";
21             var target = @"()";
22             var parser = new Parser();
23             var links = parser.Parse(source);
24             var formattedLinks = links.Format();
25             Assert.Equal(target, formattedLinks);
26         }
27
28         [Fact]
29         public static void EmptyLinkWithEmptySelfReferenceTest()
30         {
31             var source = @"(:)";
32             var parser = new Parser();
33             // '(:)' is now forbidden and should throw an exception
34             Assert.Throws<ParseException>(() => parser.Parse(source));
35         }
36
37         [Fact]
38         public static void AllFeaturesTest()
39         {
40             // Test single-line link with id
41             var input = "id: value1 value2";
42             var result = new Parser().Parse(input);
43             Assert.NotEmpty(result);
44
45             // Test multi-line link with id
46             input = "(id: value1 value2)";
47             result = new Parser().Parse(input);

```

```

48     Assert.NotEmpty(result);
49
50     // Test link without id (single-line) - now forbidden
51     input = ": value1 value2";
52     Assert.Throws<ParseException>(() => new Parser().Parse(input));
53
54     // Test link without id (multi-line) - now forbidden
55     input = "(: value1 value2)";
56     Assert.Throws<ParseException>(() => new Parser().Parse(input));
57
58     // Test singlet link
59     input = "(singlet)";
60     result = new Parser().Parse(input);
61     Assert.Single(result);
62     Assert.Null(result[0].Id);
63     Assert.NotNull(result[0].Values);
64     Assert.Single(result[0].Values!);
65     Assert.Equal("singlet", result[0].Values?[0].Id);
66     Assert.Null(result[0].Values?[0].Values);
67
68     // Test value link
69     input = "(value1 value2 value3)";
70     result = new Parser().Parse(input);
71     Assert.NotEmpty(result);
72
73     // Test quoted references
74     input = @"("id with spaces": "value with spaces")";
75     result = new Parser().Parse(input);
76     Assert.NotEmpty(result);
77
78     // Test single-quoted references
79     input = "('id': 'value')";
80     result = new Parser().Parse(input);
81     Assert.NotEmpty(result);
82
83     // Test nested links
84     input = "(outer: (inner: value))";
85     result = new Parser().Parse(input);
86     Assert.NotEmpty(result);
87 }
88
89 [Fact]
90 public static void TestSingletLinks()
91 {
92     // Test singlet (1)
93     var input = "(1)";
94     var result = new Parser().Parse(input);
95     Assert.Single(result);
96     Assert.Null(result[0].Id);
97     Assert.NotNull(result[0].Values);
98     Assert.Single(result[0].Values!);
99     Assert.Equal("1", result[0].Values?[0].Id);
100    Assert.Null(result[0].Values?[0].Values);
101
102    // Test (1 2)
103    input = "(1 2)";
104    result = new Parser().Parse(input);
105    Assert.Single(result);
106    Assert.Null(result[0].Id);
107    Assert.NotNull(result[0].Values);
108    Assert.Equal(2, result[0].Values?.Count);
109    Assert.Equal("1", result[0].Values?[0].Id);
110    Assert.Null(result[0].Values?[0].Values);
111    Assert.Equal("2", result[0].Values?[1].Id);
112    Assert.Null(result[0].Values?[1].Values);
113
114    // Test (1 2 3)
115    input = "(1 2 3)";
116    result = new Parser().Parse(input);
117    Assert.Single(result);
118    Assert.Null(result[0].Id);
119    Assert.NotNull(result[0].Values);
120    Assert.Equal(3, result[0].Values?.Count);
121    Assert.Equal("1", result[0].Values?[0].Id);
122    Assert.Null(result[0].Values?[0].Values);
123    Assert.Equal("2", result[0].Values?[1].Id);
124    Assert.Null(result[0].Values?[1].Values);
125    Assert.Equal("3", result[0].Values?[2].Id);
126    Assert.Null(result[0].Values?[2].Values);

```

```

127     // Test (1 2 3 4)
128     input = "(1 2 3 4)";
129     result = new Parser().Parse(input);
130     Assert.Single(result);
131     Assert.Null(result[0].Id);
132     Assert.NotNull(result[0].Values);
133     Assert.Equal(4, result[0].Values?.Count);
134     Assert.Equal("1", result[0].Values?[0].Id);
135     Assert.Null(result[0].Values?[0].Values);
136     Assert.Equal("2", result[0].Values?[1].Id);
137     Assert.Null(result[0].Values?[1].Values);
138     Assert.Equal("3", result[0].Values?[2].Id);
139     Assert.Null(result[0].Values?[2].Values);
140     Assert.Equal("4", result[0].Values?[3].Id);
141     Assert.Null(result[0].Values?[3].Values);
142 }
143
144 [Fact]
145 public static void EmptyDocumentTest()
146 {
147     var input = "";
148     // Empty document should return empty list
149     var result = new Parser().Parse(input);
150     Assert.Empty(result);
151 }
152
153 [Fact]
154 public static void WhitespaceOnlyTest()
155 {
156     var input = " \n \n ";
157     // Whitespace-only document should return empty list (similar to empty document)
158     var result = new Parser().Parse(input);
159     Assert.Empty(result);
160 }
161
162 [Fact]
163 public static void EmptyLinksTest()
164 {
165     var input = "()";
166     var result = new Parser().Parse(input);
167     Assert.NotEmpty(result);
168
169     // '(:)' is now forbidden
170     input = "(:)";
171     Assert.Throws<ParseException>(() => new Parser().Parse(input));
172
173     input = "(id:)";
174     result = new Parser().Parse(input);
175     Assert.NotEmpty(result);
176 }
177
178 [Fact]
179 public static void InvalidInputTest()
180 {
181     var input = "(invalid";
182     // Unclosed parentheses should throw an exception
183     Assert.Throws<ParseException>(() => new Parser().Parse(input));
184 }
185 }
186 }
187

```

1.20 ./Link.Foundation.Links.Notation.Tests/EmptyReferenceTests.cs

```

1 using System.Collections.Generic;
2 using System.Linq;
3 using Xunit;
4
5 namespace Link.Foundation.Links.Notation.Tests
6 {
7     /// <summary>
8     /// Conformance tests for the empty reference.
9     ///
10    /// https://github.com/link-foundation/links-notation/issues/288
11    ///
12    /// A bare delimiter pair is the empty reference. The three delimiters
13    /// <c>"</c>, <c>'</c> and <c>`</c> behave identically, and every longer
14    /// n-quote run keeps the meaning it already had. The table below is shared
15    /// with the Rust, JavaScript, Python, Go, Java and PHP suites, so a document
16    /// written by one implementation reads the same in all of them.

```

```

17 /// </summary>
18 public static class EmptyReferenceTests
19 {
20     /// <summary>
21     /// Renders a parsed node unambiguously: every reference is wrapped in
22     /// angle brackets so an empty one is visible as &lt;&gt;.
23     /// </summary>
24     private static string Render(Link<string> node)
25     {
26         if (node.Values == null || node.Values.Count == 0)
27         {
28             return "<" + (node.Id ?? "") + ">";
29         }
30         var head = node.Id == null ? "" : "<" + node.Id + "> ";
31         return "(" + head + string.Join(" ", node.Values.Select(Render)) + ")";
32     }
33
34     private static string Rendered(string source)
35     {
36         var links = new Parser().Parse(source);
37         return string.Join("\n", links.Select(Render));
38     }
39
40     private static void AssertParsesAs(string source, string expected)
41     {
42         Assert.Equal(expected, Rendered(source));
43     }
44
45     [Fact]
46     public static void BareDelimiterPairIsTheEmptyReference()
47     {
48         AssertParsesAs("(a \"\" b)", "(<a> <> <b>)");
49     }
50
51     [Fact]
52     public static void EveryDelimiterStyleYieldsTheSameEmptyReference()
53     {
54         AssertParsesAs("(a \"\" b)", "(<a> <> <b>)");
55         AssertParsesAs("(a '' b)", "(<a> <> <b>)");
56         AssertParsesAs("(a `` b)", "(<a> <> <b>)");
57     }
58
59     [Fact]
60     public static void AdjacentEmptyReferencesStaySeparate()
61     {
62         AssertParsesAs("(a \"\" \"\" b)", "(<a> <> <> <b>)");
63         AssertParsesAs("(a '' '' b)", "(<a> <> <> <b>)");
64         AssertParsesAs("(a `` `` b)", "(<a> <> <> <b>)");
65         AssertParsesAs("(a \"\" \"\" b)", "(<a> <> <> <b>)");
66     }
67
68     [Fact]
69     public static void NestedEmptyReferencesParse()
70     {
71         AssertParsesAs("(\"\" (\") 1)", "(<> (<> <1>))");
72         AssertParsesAs("(\"\" (' 1)", "(<> (<> <1>))");
73         AssertParsesAs("(\"\" x\" 1)", "(<x> (<> <1>))");
74         AssertParsesAs("(\"\" (\x\" 1)", "(<> (<x> <1>))");
75         AssertParsesAs("(\"\" x (\") 1)", "(<> <x> (<> <1>))");
76         AssertParsesAs("(\"\" 1 (\") 1)", "(<> <1> (<> <1>))");
77     }
78
79     [Fact]
80     public static void EmptyReferenceIsValidAsAnId()
81     {
82         AssertParsesAs("(\"\": 1)", "(<>: <1>)");
83         AssertParsesAs("(o: (\") (o: (\") 1)))", "(<o>: (<> (<o>: (<> <1>))))");
84     }
85
86     [Fact]
87     public static void NQuoteDelimitedBodiesAreUnchanged()
88     {
89         // A run that encloses a substantive body keeps its n-quote meaning.
90         AssertParsesAs("(a \"\"x\"\" b)", "(<a> <x> <b>)");
91         AssertParsesAs("(x \"\" \"\" )", "(<x> < \"\" >)");
92         AssertParsesAs("(x ' ' )", "(<x> < ' ' >)");
93         // An n-quote-delimited empty is still empty.
94         AssertParsesAs("(a \"\" \"\" b)", "(<a> <> <b>)");

```

```

95     }
96
97     [Fact]
98     public static void ASingleSpaceStillReadsAsASpace()
99     {
100         AssertParsesAs("(a \" \" b)", "<a> < > <b>");
101     }
102
103     [Fact]
104     public static void EmptyReferenceSurvivesARoundTrip()
105     {
106         var sources = new[]
107         {
108             "(a \"\" b)",
109             "(a \"\" \"\" b)",
110             "\"\" (\"\" 1))",
111             "\"\": 1)",
112             "(o: (\"\" (o: (\"\" 1))))",
113         };
114         foreach (var source in sources)
115         {
116             var formatted = ((IList<Link<string>>)new Parser().Parse(source)).Format();
117             var reparsed = ((IList<Link<string>>)new Parser().Parse(formatted)).Format();
118             Assert.Equal(formatted, reparsed);
119         }
120     }
121
122     [Fact]
123     public static void EmptyReferenceIsWrittenAsADelimiterPair()
124     {
125         var links = (IList<Link<string>>)new Parser().Parse("(a \"\" b)");
126         Assert.Equal("(a \"\" b)", links.Format());
127     }
128 }
129 }

```

1.21 ./Link.Foundation.Links.Notation.Tests/FormatConfigTests.cs

```

1  using Xunit;
2
3  namespace Link.Foundation.Links.Notation.Tests
4  {
5      public class FormatConfigTests
6      {
7          [Fact]
8          public void FormatConfigBasicTest()
9          {
10             var config = new FormatConfig();
11             Assert.False(config.LessParentheses);
12             Assert.Equal(80, config.MaxLineLength);
13             Assert.False(config.IndentLongLines);
14         }
15
16         [Fact]
17         public void FormatWithLineLengthLimitTest()
18         {
19             // Create a config with line length limit
20             // The line would be 32+ chars, so use threshold of 30
21             var config = new FormatConfig
22             {
23                 IndentLongLines = true,
24                 MaxLineLength = 30,
25                 PreferInline = false
26             };
27
28             // Verify config is set correctly
29             Assert.True(config.IndentLongLines);
30             Assert.Equal(30, config.MaxLineLength);
31             // Note: Full formatting integration would test actual output here
32         }
33
34         [Fact]
35         public void FormatWithMaxInlineRefsTest()
36         {
37             // Create a config with MaxInlineRefs=3 (should trigger indentation with 4 refs)
38             var config = new FormatConfig
39             {
40                 MaxInlineRefs = 3,
41                 PreferInline = false
42             };

```

```

43
44     // Verify config is set correctly
45     Assert.Equal(3, config.MaxInlineRefs);
46     Assert.True(config.ShouldIndentByRefCount(4));
47     // Note: Full formatting integration would test actual output here
48 }
49
50 [Fact]
51 public void FormatWithConsecutiveGroupingTest()
52 {
53     // Create a config with consecutive grouping enabled
54     var config = new FormatConfig
55     {
56         GroupConsecutive = true
57     };
58
59     // Verify config is set correctly
60     Assert.True(config.GroupConsecutive);
61     // Note: Full formatting integration would test grouping behavior here
62 }
63
64 [Fact]
65 public void FormatConfigCustomIndentTest()
66 {
67     var config = new FormatConfig
68     {
69         MaxInlineRefs = 3,
70         PreferInline = false,
71         IndentString = "    " // 4 spaces
72     };
73
74     Assert.Equal("    ", config.IndentString);
75 }
76
77 [Fact]
78 public void FormatConfigLessParenthesesTest()
79 {
80     var config = new FormatConfig
81     {
82         LessParentheses = true
83     };
84
85     Assert.True(config.LessParentheses);
86 }
87
88 [Fact]
89 public void RoundtripWithLineLengthFormattingTest()
90 {
91     // This test would require full format integration
92     // For now, just verify config creation
93     var config = new FormatConfig
94     {
95         MaxInlineRefs = 2,
96         PreferInline = false
97     };
98
99     Assert.Equal(2, config.MaxInlineRefs);
100 }
101
102 [Fact]
103 public void ShouldIndentByLengthTest()
104 {
105     var config = new FormatConfig
106     {
107         IndentLongLines = true,
108         MaxLineLength = 80
109     };
110
111     Assert.False(config.ShouldIndentByLength("short"));
112     Assert.True(config.ShouldIndentByLength(new string('a', 100)));
113 }
114
115 [Fact]
116 public void ShouldIndentByRefCountTest()
117 {
118     var config = new FormatConfig
119     {
120         MaxInlineRefs = 3
121     };
122

```

```

123         Assert.False(config.ShouldIndentByRefCount(2));
124         Assert.False(config.ShouldIndentByRefCount(3));
125         Assert.True(config.ShouldIndentByRefCount(4));
126     }
127 }
128 }

```

1.22 ./Link.Foundation.Links.Notation.Tests/IndentationConsistencyTests.cs

```

1  using Xunit;
2  using System.Collections.Generic;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public class IndentationConsistencyTests
7      {
8          [Fact]
9          public void LeadingSpacesVsNoLeadingSpaces()
10         {
11             // Example with 2 leading spaces (from issue #135)
12             var withLeading = @" TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
13 TELEGRAM ALLOWED CHATS:
14     -1002975819706
15     -1002861722681
16 TELEGRAM HIVE OVERRIDES:
17     --all-issues
18     --once
19 TELEGRAM BOT VERBOSE: true";
20
21             // Example without leading spaces (from issue #135)
22             var withoutLeading = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
23 TELEGRAM ALLOWED CHATS:
24     -1002975819706
25     -1002861722681
26 TELEGRAM HIVE OVERRIDES:
27     --all-issues
28     --once
29 TELEGRAM BOT VERBOSE: true";
30
31             var resultWith = new Parser().Parse(withLeading);
32             var resultWithout = new Parser().Parse(withoutLeading);
33
34             // Compare the entire formatted output (complete round trip test)
35             Assert.Equal(resultWithout.Format(), resultWith.Format());
36         }
37
38         [Fact]
39         public void TwoSpacesVsFourSpacesIndentation()
40         {
41             // Example with 2 spaces per level
42             var twoSpaces = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
43 TELEGRAM ALLOWED CHATS:
44     -1002975819706
45     -1002861722681
46 TELEGRAM HIVE OVERRIDES:
47     --all-issues
48     --once
49     --auto-fork
50     --skip-issues-with-prs
51     --attach-logs
52     --verbose
53     --no-tool-check
54 TELEGRAM SOLVE OVERRIDES:
55     --auto-fork
56     --auto-continue
57     --attach-logs
58     --verbose
59     --no-tool-check
60 TELEGRAM BOT VERBOSE: true";
61
62             // Example with 4 spaces per level
63             var fourSpaces = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
64 TELEGRAM ALLOWED CHATS:
65     -1002975819706
66     -1002861722681
67 TELEGRAM HIVE OVERRIDES:
68     --all-issues
69     --once
70     --auto-fork
71     --skip-issues-with-prs
72     --attach-logs
73     --verbose
74     --no-tool-check
75 TELEGRAM SOLVE OVERRIDES:

```

```

76     --auto-fork
77     --auto-continue
78     --attach-logs
79     --verbose
80     --no-tool-check
81     TELEGRAM BOT VERBOSE: true";
82
83     var resultTwo = new Parser().Parse(twoSpaces);
84     var resultFour = new Parser().Parse(fourSpaces);
85
86     // Compare the entire formatted output (complete round trip test)
87     Assert.Equal(resultFour.Format(), resultTwo.Format());
88 }
89
90 [Fact]
91 public void SimpleTwoVsFourSpacesIndentation()
92 {
93     // Simple example with 2 spaces
94     var twoSpaces = @"parent:
95 child1
96 child2";
97
98     // Simple example with 4 spaces
99     var fourSpaces = @"parent:
100 child1
101 child2";
102
103     var resultTwo = new Parser().Parse(twoSpaces);
104     var resultFour = new Parser().Parse(fourSpaces);
105
106     // Compare the entire formatted output (complete round trip test)
107     Assert.Equal(resultFour.Format(), resultTwo.Format());
108 }
109
110 [Fact]
111 public void ThreeLevelNestingWithDifferentIndentation()
112 {
113     // Three levels with 2 spaces
114     var twoSpaces = @"level1:
115 level2:
116     level3a
117     level3b
118 level2b";
119
120     // Three levels with 4 spaces
121     var fourSpaces = @"level1:
122 level2:
123     level3a
124     level3b
125 level2b";
126
127     var resultTwo = new Parser().Parse(twoSpaces);
128     var resultFour = new Parser().Parse(fourSpaces);
129
130     // Compare the entire formatted output (complete round trip test)
131     Assert.Equal(resultFour.Format(), resultTwo.Format());
132 }
133 }
134 }

```

1.23 ./Link.Foundation.Links.Notation.Tests/IndentedIdNestedValuesTests.cs

```

1  using Xunit;
2
3  namespace Link.Foundation.Links.Notation.Tests
4  {
5      public class IndentedIdNestedValuesTests
6      {
7          [Fact]
8          public void IndentedIdRetainsNestedValues()
9          {
10             var cases = new[]
11             {
12                 new[] { "outer:\n inner:\n value1\n value2\n value3", "(outer: (inner:
↪ value1 value2) value3)" },
13                 new[] { "statement:\n subject:\n I\n verb:\n love\n object:\n you:\n
↪ very\n much", "(statement: (subject: I) (verb: love) (object: (you: very much)))"
↪ },
14                 new[] { "document:\n (metadata: title author date)\n content:\n
↪ paragraph1\n (paragraph2: text (with: nested structure))\n paragraph3", "(document:
↪ (metadata: title author date) (content: paragraph1 (paragraph2: text (with: nested
↪ structure)) paragraph3))" },

```

```

15         new[] { "image:\n    file: .gitpod.Dockerfile\n    context: ./docker-content",
↪      "(image: (file: .gitpod.Dockerfile) (context: ./docker-content))" },
16         new[] { "level1:\n  level2:\n    level3a\n    level3b\n  level2b", "(level1:
↪      (level2: level3a level3b) level2b)" },
17         new[] { "root:\n  child1\n  child2\n    grandchild", "(root: child1 (child2
↪      grandchild))" },
18     };
19
20     var parser = new Parser();
21     foreach (var testCase in cases)
22     {
23         Assert.Equal(testCase[1], parser.Parse(testCase[0]).Format());
24     }
25 }
26 }
27 }

```

1.24 ./Link.Foundation.Links.Notation.Tests/IndentedIdSyntaxTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class IndentedIdSyntaxTests
7      {
8          [Fact]
9          public static void BasicIndentedIdSyntaxTest()
10         {
11             var indentedSyntax = @"3:
12     papa
13     loves
14     mama";
15
16             var inlineSyntax = "(3: papa loves mama)";
17
18             var parser = new Parser();
19             var indentedResult = parser.Parse(indentedSyntax);
20             var inlineResult = parser.Parse(inlineSyntax);
21
22             var indentedFormatted = indentedResult.Format();
23             var inlineFormatted = inlineResult.Format();
24
25             Assert.Equal(inlineFormatted, indentedFormatted);
26             Assert.Equal("(3: papa loves mama)", indentedFormatted);
27
28             [Fact]
29             public static void IndentedIdSyntaxWithSingleValueTest()
30             {
31                 var input = @"greeting:
32     hello";
33
34                 var parser = new Parser();
35                 var result = parser.Parse(input);
36                 var formatted = result.Format();
37
38                 Assert.Equal("(greeting: hello)", formatted);
39                 Assert.Single(result);
40                 Assert.Equal("greeting", result[0].Id);
41                 Assert.Single(result[0].Values!);
42                 Assert.Equal("hello", result[0].Values![0].Id);
43             }
44
45             [Fact]
46             public static void IndentedIdSyntaxWithMultipleValuesTest()
47             {
48                 var input = @"action:
49     run
50     fast
51     now";
52
53                 var parser = new Parser();
54                 var result = parser.Parse(input);
55                 var formatted = result.Format();
56
57                 Assert.Equal("(action: run fast now)", formatted);
58                 Assert.Single(result);
59                 Assert.Equal("action", result[0].Id);
60                 Assert.Equal(3, result[0].Values!.Count);
61             }
62

```

```

63     [Fact]
64     public static void IndentedIdSyntaxWithNumericIdTest()
65     {
66         var input = @"42:
67 answer
68 to
69 everything";
70
71         var parser = new Parser();
72         var result = parser.Parse(input);
73         var formatted = result.Format();
74
75         Assert.Equal("(42: answer to everything)", formatted);
76     }
77
78     [Fact]
79     public static void IndentedIdSyntaxWithQuotedIdTest()
80     {
81         var input = @"""complex id"":
82 value1
83 value2";
84
85         var parser = new Parser();
86         var result = parser.Parse(input);
87         var formatted = result.Format();
88
89         Assert.Equal("( 'complex id': value1 value2)", formatted);
90     }
91
92     [Fact]
93     public static void MultipleIndentedIdLinksTest()
94     {
95         var input = @"first:
96 a
97 b
98 second:
99 c
100 d";
101
102         var parser = new Parser();
103         var result = parser.Parse(input);
104         var formatted = result.Format();
105
106         Assert.Equal(2, result.Count);
107         Assert.Contains("(first: a b)", formatted);
108         Assert.Contains("(second: c d)", formatted);
109     }
110
111     [Fact]
112     public static void MixedIndentedAndRegularSyntaxTest()
113     {
114         var input = @"first:
115 a
116 b
117 (second: c d)
118 third value";
119
120         var parser = new Parser();
121         var result = parser.Parse(input);
122
123         Assert.Equal(3, result.Count);
124
125         var formatted = result.Format();
126         Assert.Contains("(first: a b)", formatted);
127         Assert.Contains("(second: c d)", formatted);
128         Assert.Contains("third value", formatted);
129     }
130
131     [Fact]
132     public static void UnsupportedColonOnlySyntaxShouldFailTest()
133     {
134         var input = @"
135 papa
136 loves
137 mama";
138
139         var parser = new Parser();
140         Assert.Throws<ParseException>(() => parser.Parse(input));
141     }
142
143     [Fact]

```

```

144 public static void EmptyIndentedIdShouldWorkTest()
145 {
146     var input = "empty: ";
147
148     var parser = new Parser();
149     var result = parser.Parse(input);
150
151     Assert.Single(result);
152     Assert.Equal("empty", result[0].Id);
153     var emptyValues = result[0].Values;
154     Assert.True(emptyValues == null || emptyValues.Count == 0);
155
156     var formatted = result.Format();
157     Assert.Equal("(empty)", formatted);
158 }
159
160 [Fact]
161 public static void EquivalenceTestComprehensiveTest()
162 {
163     var testCases = new[]
164     {
165         new { Indented = "test:\n one", Inline = "(test: one)" },
166         new { Indented = "x:\n a\n b\n c", Inline = "(x: a b c)" },
167         new { Indented = "\"quoted\":\n value", Inline = "(\"quoted\": value)" }
168     };
169
170     var parser = new Parser();
171
172     foreach (var testCase in testCases)
173     {
174         var indentedResult = parser.Parse(testCase.Indented);
175         var inlineResult = parser.Parse(testCase.Inline);
176
177         var indentedFormatted = indentedResult.Format();
178         var inlineFormatted = inlineResult.Format();
179
180         Assert.Equal(inlineFormatted, indentedFormatted);
181     }
182 }
183
184 [Fact]
185 public static void IndentedIdWithDeeperNestingTest()
186 {
187     var input = @"root:
188 child1
189 child2
190 grandchild";
191
192     var parser = new Parser();
193     var result = parser.Parse(input);
194
195     Assert.NotEmpty(result);
196
197     var rootLink = result[0];
198     Assert.Equal("root", rootLink.Id);
199     Assert.Equal(2, rootLink.Values!.Count);
200     Assert.Equal("(root: child1 (child2 grandchild))", result.Format());
201 }
202 }
203 }

```

1.25 ./Link.Foundation.Links.Notation.Tests/LinkTests.cs

```

1 using System;
2 using Xunit;
3 using System.Collections.Generic;
4
5 namespace Link.Foundation.Links.Notation.Tests
6 {
7     public static class LinkTests
8     {
9         [Fact]
10        public static void LinkConstructorWithIdOnlyTest()
11        {
12            var link = new Link<string>("test", null);
13            Assert.Equal("test", link.Id);
14            Assert.Null(link.Values);
15        }
16
17        [Fact]
18        public static void LinkConstructorWithIdAndValuesTest()

```

```

19     {
20         var values = new List<Link<string>> { new Link<string>("value1"), new
↪ Link<string>("value2") };
21         var link = new Link<string>("parent", values);
22         Assert.Equal("parent", link.Id);
23         Assert.Equal(2, link.Values?.Count);
24     }
25
26     [Fact]
27     public static void LinkToStringWithIdOnlyTest()
28     {
29         var link = new Link<string>("test");
30         Assert.Equal("(test)", link.ToString());
31     }
32
33     [Fact]
34     public static void LinkToStringWithValuesOnlyTest()
35     {
36         var values = new List<Link<string>> { new Link<string>("value1"), new
↪ Link<string>("value2") };
37         var link = new Link<string>(null, values);
38         Assert.Equal("(value1 value2)", link.ToString());
39     }
40
41     [Fact]
42     public static void LinkToStringWithIdAndValuesTest()
43     {
44         var values = new List<Link<string>> { new Link<string>("child1"), new
↪ Link<string>("child2") };
45         var link = new Link<string>("parent", values);
46         Assert.Equal("(parent: child1 child2)", link.ToString());
47     }
48
49     [Fact]
50     public static void LinkEqualsTest()
51     {
52         var link1 = new Link<string>("test");
53         var link2 = new Link<string>("test");
54         var link3 = new Link<string>("other");
55
56         Assert.Equal(link1, link2);
57         Assert.NotEqual(link1, link3);
58     }
59
60     [Fact]
61     public static void LinkCombineTest()
62     {
63         var link1 = new Link<string>("first");
64         var link2 = new Link<string>("second");
65         var combined = link1.Combine(link2);
66
67         Assert.Null(combined.Id);
68         Assert.Equal(2, combined.Values?.Count);
69         Assert.Equal("first", combined.Values?[0].Id);
70         Assert.Equal("second", combined.Values?[1].Id);
71     }
72
73     [Fact]
74     public static void LinkEscapeReferenceSimpleTest()
75     {
76         Assert.Equal("test", Link<string>.EscapeReference("test"));
77         Assert.Equal("test123", Link<string>.EscapeReference("test123"));
78     }
79
80     [Fact]
81     public static void LinkEscapeReferenceWithSpecialCharactersTest()
82     {
83         Assert.Equal("'has space'", Link<string>.EscapeReference("has space"));
84         Assert.Equal("'has:colon'", Link<string>.EscapeReference("has:colon"));
85         Assert.Equal("'has(paren)'", Link<string>.EscapeReference("has(paren)"));
86         Assert.Equal("'has)paren'", Link<string>.EscapeReference("has)paren"));
87         Assert.Equal("'\"has'quote'", Link<string>.EscapeReference("has'quote"));
88         Assert.Equal("'has\\\"quote'", Link<string>.EscapeReference("has\"quote"));
89     }
90
91     [Fact]
92     public static void LinkSimplifyTest()
93     {

```

```

94         // Test simplify with empty values
95         var link1 = new Link<string>("test", null);
96         var simplified1 = link1.Simplify();
97         Assert.Equal(link1, simplified1);
98
99         // Test simplify with single value
100        var link2 = new Link<string>(null, new List<Link<string>> { new
↪ Link<string>("single", null) });
101        var simplified2 = link2.Simplify();
102        Assert.Equal("single", simplified2.Id);
103        Assert.Null(simplified2.Values);
104    }
105 }
106 }

```

1.26 ./Link.Foundation.Links.Notation.Tests/LinksGroupTests.cs

```

1  using System;
2  using System.Collections.Generic;
3  using Xunit;
4
5  namespace Link.Foundation.Links.Notation.Tests
6  {
7      public static class LinksGroupTests
8      {
9          [Fact]
10         public static void LinksGroupConstructorTest()
11         {
12             var element = new Link<string>("root", null);
13             var groups = new List<LinksGroup<string>>
14             {
15                 new LinksGroup<string>(new Link<string>("child1", null)),
16                 new LinksGroup<string>(new Link<string>("child2", null))
17             };
18             var group = new LinksGroup<string>(element, groups);
19
20             Assert.Equal(element, group.Link);
21             Assert.Equal(groups, group.Groups);
22         }
23
24         [Fact]
25         public static void LinksGroupToListFlattensStructureTest()
26         {
27             var root = new Link<string>("root", null);
28             var child1 = new Link<string>("child1", null);
29             var child2 = new Link<string>("child2", null);
30             var grandchild = new Link<string>("grandchild", null);
31
32             var childGroup = new LinksGroup<string>(child2, new List<LinksGroup<string>>
33             {
34                 new LinksGroup<string>(grandchild)
35             });
36
37             var group = new LinksGroup<string>(root, new List<LinksGroup<string>>
38             {
39                 new LinksGroup<string>(child1),
40                 childGroup
41             });
42
43             var list = group.ToLinksList();
44             // The C# implementation creates a hierarchical structure
45             // root, root.Combine(child1), root.Combine(child2),
↪ root.Combine(child2).Combine(grandchild)
46             Assert.Equal(4, list.Count);
47             Assert.Equal(root, list[0]);
48         }
49
50         [Fact]
51         public static void LinksGroupAppendToLinksListTest()
52         {
53             var element = new Link<string>("root", null);
54             var children = new List<LinksGroup<string>>
55             {
56                 new LinksGroup<string>(new Link<string>("child1", null)),
57                 new LinksGroup<string>(new Link<string>("child2", null))
58             };
59             var group = new LinksGroup<string>(element, children);
60
61             var list = new List<Link<string>>();
62             group.AppendToLinksList(list);

```

```

63     Assert.Equal(3, list.Count);
64     Assert.Equal(element, list[0]);
65 }
66
67 [Fact]
68 public static void LinksGroupToStringTest()
69 {
70     var element = new Link<string>("root", null);
71     var children = new List<LinksGroup<string>>
72     {
73         new LinksGroup<string>(new Link<string>("child1", null)),
74         new LinksGroup<string>(new Link<string>("child2", null))
75     };
76     var group = new LinksGroup<string>(element, children);
77
78     var str = group.ToString();
79     Assert.Contains("root", str);
80     Assert.Contains("child1", str);
81     Assert.Contains("child2", str);
82 }
83
84 [Fact]
85 public static void LinksGroupConstructorEquivalentTest()
86 {
87     // Test creating a LinksGroup structure in an equivalent way
88     var root = new Link<string>("root", null);
89     var children = new List<Link<string>>
90     {
91         new Link<string>("child1", null),
92         new Link<string>("child2", null)
93     };
94
95     // Create a group with an id
96     var groupElement = new Link<string>("group", null);
97     var childGroups = new List<LinksGroup<string>>
98     {
99         new LinksGroup<string>(root),
100         new LinksGroup<string>(children[0]),
101         new LinksGroup<string>(children[1])
102     };
103     var group = new LinksGroup<string>(groupElement, childGroups);
104
105     Assert.Equal("group", group.Link.Id);
106     Assert.Equal(3, group.Groups!.Count);
107     Assert.Equal(root, group.Groups[0].Link);
108 }
109 }
110 }
111 }

```

1.27 ./Link.Foundation.Links.Notation.Tests/MixedIndentationModesTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MixedIndentationModesTests
7      {
8          [Fact]
9          public static void HeroExampleMixedModesTest()
10         {
11             var input = @"empInfo
12 employees:
13 (
14     name (James Kirk)
15     age 40
16 )
17 (
18     name (Jean-Luc Picard)
19     age 45
20 )
21 (
22     name (Wesley Crusher)
23     age 27
24 )";
25
26             var parser = new Parser();
27             var result = parser.Parse(input);
28

```

```

29         Assert.NotEmpty(result);
30         var formatted = result.Format();
31         Assert.Contains("empInfo", formatted);
32         Assert.Contains("employees", formatted);
33         Assert.Contains("James Kirk", formatted);
34         Assert.Contains("Jean-Luc Picard", formatted);
35         Assert.Contains("Wesley Crusher", formatted);
36     }
37
38     [Fact]
39     public static void HeroExampleAlternativeFormatTest()
40     {
41         var input = @"empInfo
42 (
43     employees:
44     (
45         name (James Kirk)
46         age 40
47     )
48     (
49         name (Jean-Luc Picard)
50         age 45
51     )
52     (
53         name (Wesley Crusher)
54         age 27
55     )
56 )";
57
58         var parser = new Parser();
59         var result = parser.Parse(input);
60
61         Assert.NotEmpty(result);
62         var formatted = result.Format();
63         Assert.Contains("empInfo", formatted);
64         Assert.Contains("employees:", formatted);
65         Assert.Contains("James Kirk", formatted);
66         Assert.Contains("Jean-Luc Picard", formatted);
67         Assert.Contains("Wesley Crusher", formatted);
68     }
69
70     [Fact]
71     public static void HeroExampleEquivalenceTest()
72     {
73         var version1 = @"empInfo
74 employees:
75 (
76     name (James Kirk)
77     age 40
78 )
79 (
80     name (Jean-Luc Picard)
81     age 45
82 )
83 (
84     name (Wesley Crusher)
85     age 27
86 )";
87
88         var version2 = @"empInfo
89 (
90     employees:
91     (
92         name (James Kirk)
93         age 40
94     )
95     (
96         name (Jean-Luc Picard)
97         age 45
98     )
99     (
100        name (Wesley Crusher)
101        age 27
102    )
103 )";
104
105         var parser = new Parser();
106         var result1 = parser.Parse(version1);
107         var result2 = parser.Parse(version2);

```

```

108         Assert.NotEmpty(result1);
109         Assert.NotEmpty(result2);
110
111         var formatted1 = result1.Format();
112         var formatted2 = result2.Format();
113
114         Assert.Equal(formatted1, formatted2);
115     }
116
117     [Fact]
118     public static void SetContextWithoutColonTest()
119     {
120         var input = @"empInfo
121 employees";
122
123         var parser = new Parser();
124         var result = parser.Parse(input);
125
126         Assert.NotEmpty(result);
127         var formatted = result.Format();
128         Assert.Contains("empInfo", formatted);
129         Assert.Contains("employees", formatted);
130     }
131
132     [Fact]
133     public static void SequenceContextWithColonTest()
134     {
135         var input = @"employees:
136 James Kirk
137 Jean-Luc Picard
138 Wesley Crusher";
139
140         var parser = new Parser();
141         var result = parser.Parse(input);
142
143         Assert.NotEmpty(result);
144         Assert.Single(result);
145         var formatted = result.Format();
146         Assert.Contains("employees:", formatted);
147         Assert.Contains("James Kirk", formatted);
148         Assert.Contains("Jean-Luc Picard", formatted);
149         Assert.Contains("Wesley Crusher", formatted);
150     }
151
152     [Fact]
153     public static void SequenceContextWithComplexValuesTest()
154     {
155         var input = @"employees:
156 (
157     name (James Kirk)
158     age 40
159 )
160 (
161     name (Jean-Luc Picard)
162     age 45
163 )"
164         ;
165
166         var parser = new Parser();
167         var result = parser.Parse(input);
168
169         Assert.NotEmpty(result);
170         Assert.Single(result);
171         var formatted = result.Format();
172         Assert.Contains("employees:", formatted);
173         Assert.Contains("James Kirk", formatted);
174         Assert.Contains("Jean-Luc Picard", formatted);
175     }
176
177     [Fact]
178     public static void NestedSetAndSequenceContextsTest()
179     {
180         var input = @"company
181 departments:
182     engineering
183     sales
184 employees:
185     (name John)
186     (name Jane)";
187

```

```

188         var parser = new Parser();
189         var result = parser.Parse(input);
190
191         Assert.NotEmpty(result);
192         var formatted = result.Format();
193         Assert.Contains("company", formatted);
194         Assert.Contains("departments:", formatted);
195         Assert.Contains("employees:", formatted);
196     }
197
198     [Fact]
199     public static void DeeplyNestedMixedModesTest()
200     {
201         var input = @"root
202 level1
203   level2:
204     value1
205     value2
206   level2b
207     level3";
208
209         var parser = new Parser();
210         var result = parser.Parse(input);
211
212         Assert.NotEmpty(result);
213         var formatted = result.Format();
214         Assert.Contains("root", formatted);
215         Assert.Contains("level2:", formatted);
216     }
217 }
218 }

```

1.28 ./Link.Foundation.Links.Notation.Tests/MultiQuoteParserTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MultiQuoteParserTests
7      {
8          // Helper to extract single reference ID
9          private static string? GetSingleRefId(System.Collections.Generic.IList<Link<string>>
10 ↪ result)
11          {
12              if (result.Count == 1 && result[0].Id == null && result[0].Values?.Count == 1)
13              {
14                  return result[0].Values![0].Id;
15              }
16              return result.Count == 1 ? result[0].Id : null;
17          }
18
19          // =====
20          // Backtick Quote Tests (Single Backtick)
21          // =====
22
23          [Fact]
24          public static void TestBacktickQuotedReference()
25          {
26              var parser = new Parser();
27              var result = parser.Parse("`backtick quoted`");
28              Assert.Equal("backtick quoted", GetSingleRefId(result));
29          }
30
31          [Fact]
32          public static void TestBacktickQuotedWithSpaces()
33          {
34              var parser = new Parser();
35              var result = parser.Parse("`text with spaces`");
36              Assert.Equal("text with spaces", GetSingleRefId(result));
37          }
38
39          [Fact]
40          public static void TestBacktickQuotedMultiline()
41          {
42              var parser = new Parser();
43              var result = parser.Parse("`line1\nline2`");
44              Assert.Single(result);
45              Assert.NotNull(result[0].Values);
46              Assert.Single(result[0].Values!);

```

```

46     Assert.Equal("line1\nline2", result[0].Values![0].Id);
47 }
48
49 [Fact]
50 public static void TestBacktickQuotedWithEscapedBacktick()
51 {
52     var parser = new Parser();
53     var result = parser.Parse("`text with `` escaped backtick`");
54     Assert.Equal("text with ` escaped backtick", GetSingleRefId(result));
55 }
56
57 // =====
58 // Single Quote Tests (with escaping)
59 // =====
60
61 [Fact]
62 public static void TestSingleQuoteWithEscapedSingleQuote()
63 {
64     var parser = new Parser();
65     var result = parser.Parse("'text with ' escaped quote'");
66     Assert.Equal("text with ' escaped quote", GetSingleRefId(result));
67 }
68
69 // =====
70 // Double Quote Tests (with escaping)
71 // =====
72
73 [Fact]
74 public static void TestDoubleQuoteWithEscapedDoubleQuote()
75 {
76     var parser = new Parser();
77     var result = parser.Parse("\"text with \" escaped quote\"");
78     Assert.Equal("text with \" escaped quote", GetSingleRefId(result));
79 }
80
81 // =====
82 // Double Quotes (2 quote chars) Tests
83 // =====
84
85 [Fact]
86 public static void TestDoubleDoubleQuotes()
87 {
88     var parser = new Parser();
89     var result = parser.Parse("\"\"double double quotes\"");
90     Assert.Equal("double double quotes", GetSingleRefId(result));
91 }
92
93 [Fact]
94 public static void TestDoubleDoubleQuotesWithSingleQuoteInside()
95 {
96     var parser = new Parser();
97     var result = parser.Parse("\"\"text with \" inside\"");
98     Assert.Equal("text with \" inside", GetSingleRefId(result));
99 }
100
101 [Fact]
102 public static void TestDoubleDoubleQuotesWithEscape()
103 {
104     var parser = new Parser();
105     var result = parser.Parse("\"\"text with \"\"\" escaped double\"");
106     Assert.Equal("text with \"\" escaped double", GetSingleRefId(result));
107 }
108
109 [Fact]
110 public static void TestDoubleSingleQuotes()
111 {
112     var parser = new Parser();
113     var result = parser.Parse("'double single quotes'");
114     Assert.Equal("double single quotes", GetSingleRefId(result));
115 }
116
117 [Fact]
118 public static void TestDoubleSingleQuotesWithSingleQuoteInside()
119 {
120     var parser = new Parser();
121     var result = parser.Parse("'text with ' inside'");
122     Assert.Equal("text with ' inside", GetSingleRefId(result));
123 }
124

```

```

125 [Fact]
126 public static void TestDoubleSingleQuotesWithEscape()
127 {
128     var parser = new Parser();
129     var result = parser.Parse("'text with ''' escaped single'");
130     Assert.Equal("text with ' escaped single", GetSingleRefId(result));
131 }
132
133 [Fact]
134 public static void TestDoubleBacktickQuotes()
135 {
136     var parser = new Parser();
137     var result = parser.Parse("`double backtick quotes`");
138     Assert.Equal("double backtick quotes", GetSingleRefId(result));
139 }
140
141 [Fact]
142 public static void TestDoubleBacktickQuotesWithBacktickInside()
143 {
144     var parser = new Parser();
145     var result = parser.Parse("`text with ` inside`");
146     Assert.Equal("text with ` inside", GetSingleRefId(result));
147 }
148
149 [Fact]
150 public static void TestDoubleBacktickQuotesWithEscape()
151 {
152     var parser = new Parser();
153     var result = parser.Parse("`text with `` escaped backtick`");
154     Assert.Equal("text with `` escaped backtick", GetSingleRefId(result));
155 }
156
157 // =====
158 // Triple Quotes (3 quote chars) Tests
159 // =====
160
161 [Fact]
162 public static void TestTripleDoubleQuotes()
163 {
164     var parser = new Parser();
165     var result = parser.Parse("\"\"\"triple double quotes\"\"\"");
166     Assert.Equal("triple double quotes", GetSingleRefId(result));
167 }
168
169 [Fact]
170 public static void TestTripleDoubleQuotesWithDoubleQuoteInside()
171 {
172     var parser = new Parser();
173     var result = parser.Parse("\"\"\"text with \"\" inside\"\"\"");
174     Assert.Equal("text with \"\" inside", GetSingleRefId(result));
175 }
176
177 [Fact]
178 public static void TestTripleDoubleQuotesWithEscape()
179 {
180     var parser = new Parser();
181     var result = parser.Parse("\"\"\"text with \"\"\"\"\" escaped triple\"\"\"");
182     Assert.Equal("text with \"\"\"\"\" escaped triple", GetSingleRefId(result));
183 }
184
185 [Fact]
186 public static void TestTripleSingleQuotes()
187 {
188     var parser = new Parser();
189     var result = parser.Parse(''''triple single quotes''');
190     Assert.Equal("triple single quotes", GetSingleRefId(result));
191 }
192
193 [Fact]
194 public static void TestTripleBacktickQuotes()
195 {
196     var parser = new Parser();
197     var result = parser.Parse("`triple backtick quotes`");
198     Assert.Equal("triple backtick quotes", GetSingleRefId(result));
199 }
200
201 // =====
202 // Quadruple Quotes (4 quote chars) Tests

```

```

203 // =====
204
205 [Fact]
206 public static void TestQuadrupleDoubleQuotes()
207 {
208     var parser = new Parser();
209     var result = parser.Parse("\\""\\""\\""\\"quadruple double quotes\\""\\""\\"");
210     Assert.Equal("quadruple double quotes", GetSingleRefId(result));
211 }
212
213 [Fact]
214 public static void TestQuadrupleSingleQuotes()
215 {
216     var parser = new Parser();
217     var result = parser.Parse("''''quadruple single quotes''''");
218     Assert.Equal("quadruple single quotes", GetSingleRefId(result));
219 }
220
221 [Fact]
222 public static void TestQuadrupleBacktickQuotes()
223 {
224     var parser = new Parser();
225     var result = parser.Parse("````quadruple backtick quotes````");
226     Assert.Equal("quadruple backtick quotes", GetSingleRefId(result));
227 }
228
229 // =====
230 // Quintuple Quotes (5 quote chars) Tests
231 // =====
232
233 [Fact]
234 public static void TestQuintupleDoubleQuotes()
235 {
236     var parser = new Parser();
237     var result = parser.Parse("\\""\\""\\""\\""\\"quintuple double quotes\\""\\""\\""\\"");
238     Assert.Equal("quintuple double quotes", GetSingleRefId(result));
239 }
240
241 [Fact]
242 public static void TestQuintupleSingleQuotes()
243 {
244     var parser = new Parser();
245     var result = parser.Parse("'''''quintuple single quotes'''''");
246     Assert.Equal("quintuple single quotes", GetSingleRefId(result));
247 }
248
249 [Fact]
250 public static void TestQuintupleBacktickQuotes()
251 {
252     var parser = new Parser();
253     var result = parser.Parse("``````quintuple backtick quotes``````");
254     Assert.Equal("quintuple backtick quotes", GetSingleRefId(result));
255 }
256
257 // =====
258 // Complex Scenarios Tests
259 // =====
260
261 [Fact]
262 public static void TestMixedQuotesInLink()
263 {
264     var parser = new Parser();
265     var result = parser.Parse("(\"double\" 'single' `backtick`)");
266     Assert.Single(result);
267     Assert.NotNull(result[0].Values);
268     Assert.Equal(3, result[0].Values!.Count);
269     Assert.Equal("double", result[0].Values![0].Id);
270     Assert.Equal("single", result[0].Values![1].Id);
271     Assert.Equal("backtick", result[0].Values![2].Id);
272 }
273
274 [Fact]
275 public static void TestBacktickAsIdInLink()
276 {
277     var parser = new Parser();
278     var result = parser.Parse("`myId` : value1 value2");
279     Assert.Single(result);
280     Assert.Equal("myId", result[0].Id);

```

```

281     Assert.NotNull(result[0].Values);
282     Assert.Equal(2, result[0].Values!.Count);
283 }
284
285 [Fact]
286 public static void TestCodeBlockLikeContent()
287 {
288     var parser = new Parser();
289     var result = parser.Parse("```const x = 1;```");
290     Assert.Equal("const x = 1;", GetSingleRefId(result));
291 }
292
293 [Fact]
294 public static void TestNestedQuotesInMarkdown()
295 {
296     var parser = new Parser();
297     var result = parser.Parse("`Use `code` in markdown`");
298     Assert.Equal("Use `code` in markdown", GetSingleRefId(result));
299 }
300
301 [Fact]
302 public static void TestJsonStringWithQuotes()
303 {
304     var parser = new Parser();
305     var result = parser.Parse("\"{ \"key\": \"value\"}\"");
306     Assert.Equal("{ \"key\": \"value\"}", GetSingleRefId(result));
307 }
308
309 // =====
310 // Edge Cases
311 // =====
312
313 [Fact]
314 public static void TestWhitespacePreservedInQuotes()
315 {
316     var parser = new Parser();
317     var result = parser.Parse("\" spaces \"");
318     Assert.Equal(" spaces ", GetSingleRefId(result));
319 }
320
321 [Fact]
322 public static void TestMultilineInDoubleDoubleQuotes()
323 {
324     var parser = new Parser();
325     var result = parser.Parse("(\"line1\nline2\")");
326     Assert.Single(result);
327     Assert.NotNull(result[0].Values);
328     Assert.Single(result[0].Values!);
329     Assert.Equal("line1\nline2", result[0].Values![0].Id);
330 }
331
332 // =====
333 // Unlimited Quotes (6+ quote chars) Tests
334 // =====
335
336 [Fact]
337 public static void TestUnlimitedQuotes6()
338 {
339     // Test 6-quote strings
340     var parser = new Parser();
341     var result = parser.Parse("\"\"\"\"hello\"\"\"");
342     Assert.Equal("hello", GetSingleRefId(result));
343 }
344
345 [Fact]
346 public static void TestUnlimitedQuotes10()
347 {
348     // Test 10-quote strings
349     var parser = new Parser();
350     var result = parser.Parse("\"\"\"\"\"\"very deeply
351     ↪ quoted\"\"\"\"\"");
352     Assert.Equal("very deeply quoted", GetSingleRefId(result));
353 }
354
355 [Fact]
356 public static void TestUnlimitedQuotes6WithInnerQuotes()
357 {
358     // Test 6-quote strings with inner 5-quote sequences

```

```

358         var parser = new Parser();
359         var result = parser.Parse(@"\"\"\"\"\"hello with \"\"\"\"\" five quotes
↪ inside\"\"\"\"\"");
        Assert.Equal("hello with \"\"\"\"\" five quotes inside", GetSingleRefId(result));
    }

    [Fact]
    public static void TestUnlimitedSingleQuotes7()
    {
        // Test 7-quote single quote strings
        var parser = new Parser();
        var result = parser.Parse("''''''seven single quotes''''''");
        Assert.Equal("seven single quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestUnlimitedBackticks8()
    {
        // Test 8-quote backtick strings
        var parser = new Parser();
        var result = parser.Parse("`````````eight backticks````````");
        Assert.Equal("eight backticks", GetSingleRefId(result));
    }
}

```

1.29 ./Link.Foundation.Links.Notation.Tests/MultilineParserTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MultilineParserTests
7      {
8          [Fact]
9          public static void TwoLinksTest()
10         {
11             var source = @"(first: x y
12 (second: a b)";
13             var parser = new Parser();
14             var links = parser.Parse(source);
15             var target = links.Format();
16             Assert.Equal(source, target);
17         }
18
19         [Fact]
20         public static void ParseAndStringifyTest()
21         {
22             var source = @"(papa (lovesMama: loves mama))
23 (son lovesMama)
24 (daughter lovesMama)
25 (all (love mama))";
26             var parser = new Parser();
27             var links = parser.Parse(source);
28             var target = links.Format();
29             Assert.Equal(source, target);
30         }
31
32         [Fact]
33         public static void ParseAndStringifyTest2()
34         {
35             var source = @"father (lovesMom: loves mom)
36 son lovesMom
37 daughter lovesMom
38 all (love mom)";
39             var links = new Parser().Parse(source);
40             var target = links.Format(lessParentheses: true);
41             Assert.Equal(source, target);
42         }
43
44         [Fact]
45         public static void ParseAndStringifyWithLessParenthesesTest()
46         {
47             var source = @"lovesMama: loves mama
48 papa lovesMama
49 son lovesMama
50 daughter lovesMama
51 all (love mama)";
52             var parser = new Parser();

```

```

53         var links = parser.Parse(source);
54         var target = links.Format(lessParentheses: true);
55         Assert.Equal(source, target);
56     }
57
58     [Fact]
59     public static void DuplicateIdentifiersTest()
60     {
61         var source = @"(a: a b)
62 (a: b c)";
63         var target = @"(a: a b)
64 (a: b c)";
65         var parser = new Parser();
66         var links = parser.Parse(source);
67         var formattedLinks = links.Format();
68         Assert.Equal(target, formattedLinks);
69     }
70
71     [Fact]
72     public static void ComplexStructureTest()
73     {
74         var input = @"(Type: Type Type)
75 Number
76 String
77 Array
78 Value
79 (property: name type)
80 (method: name params return)";
81
82         var result = new Parser().Parse(input);
83         Assert.NotEmpty(result);
84     }
85
86     [Fact]
87     public static void MixedFormatsTest()
88     {
89         // Mix of single-line and multi-line formats
90         var input = @"id1: value1
91 (id2: value2 value3)
92 simple ref
93 (complex:
94     nested1
95     nested2
96 )";
97
98         var result = new Parser().Parse(input);
99         Assert.NotEmpty(result);
100     }
101
102     [Fact]
103     public static void MultilineWithIdTest()
104     {
105         // Test multi-line link with id
106         var input = "(id: value1 value2)";
107         var result = new Parser().Parse(input);
108         Assert.NotEmpty(result);
109     }
110
111     [Fact]
112     public static void MultipleTopLevelElementsTest()
113     {
114         // Test multiple top-level elements
115         var input = "(elem1: val1)\n(elem2: val2)";
116         var result = new Parser().Parse(input);
117         Assert.NotEmpty(result);
118     }
119
120     [Fact]
121     public static void MultilineSimpleLinksTest()
122     {
123         var input = "(1: 1 1)\n(2: 2 2)";
124         var parser = new Parser();
125         var parsed = parser.Parse(input);
126         Assert.NotEmpty(parsed);
127
128         // Validate regular formatting
129         var output = parsed.Format();
130         Assert.Contains("(1: 1 1)", output);
131         Assert.Contains("(2: 2 2)", output);

```

```

132
133         // Validate alternate formatting matches input
134         var outputAlternate = parsed.Format();
135         Assert.Equal(input, outputAlternate);
136     }
137
138     [Fact]
139     public static void IndentedChildrenTest()
140     {
141         var input = "parent\n  child1\n  child2";
142         var parser = new Parser();
143         var parsed = parser.Parse(input);
144
145         // The parsed structure should have parent with children
146         Assert.NotEmpty(parsed);
147     }
148 }
149 }

```

1.30 ./Link.Foundation.Links.Notation.Tests/MultilineQuotedStringTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MultilineQuotedStringTests
7      {
8          [Fact]
9          public static void TestMultilineDoubleQuotedReference()
10         {
11             var input = @"(
12                 ""long
13                 string literal representing
14                 the reference""
15
16                 'another
17                 long string literal
18                 as another reference'
19             )";
20
21             var parser = new Parser();
22             var result = parser.Parse(input);
23
24             Assert.NotEmpty(result);
25             Assert.Single(result);
26
27             var link = result[0];
28             Assert.Null(link.Id);
29             Assert.NotNull(link.Values);
30             Assert.Equal(2, link.Values.Count);
31
32             // The two references sit on separate lines inside the parentheses, so the
33             // nested context turns each of them into its own link.
34             Assert.Equal(@"long
35 string literal representing
36 the reference", link.Values![0].Values![0].Id);
37
38             Assert.Equal(@"another
39 long string literal
40 as another reference", link.Values![1].Values![0].Id);
41         }
42
43         [Fact]
44         public static void TestSimpleMultilineDoubleQuoted()
45         {
46             var input = @"("""line1
47 line2""")";
48             var parser = new Parser();
49             var result = parser.Parse(input);
50
51             Assert.NotEmpty(result);
52             Assert.Single(result);
53
54             var link = result[0];
55             Assert.Null(link.Id);
56             Assert.NotNull(link.Values);
57             Assert.Single(link.Values);
58             Assert.Equal("line1\nline2", link.Values[0].Id);
59         }
60
61         [Fact]

```

```

61     public static void TestSimpleMultilineSingleQuoted()
62     {
63         var input = @"('line1
64 line2')";
65         var parser = new Parser();
66         var result = parser.Parse(input);
67
68         Assert.NotEmpty(result);
69         Assert.Single(result);
70
71         var link = result[0];
72         Assert.Null(link.Id);
73         Assert.NotNull(link.Values);
74         Assert.Single(link.Values);
75         Assert.Equal("line1\nline2", link.Values[0].Id);
76     }
77
78     [Fact]
79     public static void TestMultilineQuotedAsId()
80     {
81         var input = @"("multi
82 line
83 id": value1 value2)";
84         var parser = new Parser();
85         var result = parser.Parse(input);
86
87         Assert.NotEmpty(result);
88         Assert.Single(result);
89
90         var link = result[0];
91         Assert.Equal("multi\nline\nid", link.Id);
92         Assert.NotNull(link.Values);
93         Assert.Equal(2, link.Values.Count);
94     }
95 }
96 }

```

1.31 ./Link.Foundation.Links.Notation.Tests/NestedIndentationTests.cs

```

1  using Xunit;
2
3  namespace Link.Foundation.Links.Notation.Tests
4  {
5      /// <summary>
6      /// Tests for indentation inside parentheses.
7      ///
8      /// See https://github.com/link-foundation/links-notation/issues/282
9      ///
10     /// Indentation is structural at the root, so it must be structural inside parentheses too:
11     /// a parenthesized group opens a nested context that starts fresh at indentation level zero
12     /// and follows exactly the root's rules.
13     /// </summary>
14     public class NestedIndentationTests
15     {
16         private static string FormatSource(string source) => new Parser().Parse(source).Format();
17
18         private static void AssertFormat(string source, string expected) =>
19         ↪ Assert.Equal(expected, FormatSource(source));
20
21         [Fact]
22         public void ParenthesesReproduceRootIndentation()
23         {
24             AssertFormat("a\n b\nc\n d", "(a)\n((a) (b))\n(c)\n((c) (d))");
25
26             // The same lines inside parentheses keep the same structure, nested under
27             // the link the group belongs to.
28             ↪ AssertFormat("array (\n a\n b\n c\n d\n)", "(array ((a) ((a) (b)) (c) ((c) (d))))");
29         }
30
31         [Fact]
32         public void ParenthesesKeepRecordBoundaries()
33         {
34             var source = "value (\n id \"1\"\n label \"one\"\n)";
35             AssertFormat(source, "(value ((id 1) (label one)))");
36
37             var links = new Parser().Parse(source);
38             Assert.Single(links);
39
40             var group = links[0].Values![1];

```

```

40     Assert.Null(group.Id);
41     Assert.Equal(2, group.Values!.Count);
42     Assert.Equal("id", group.Values[0].Values![0].Id);
43     Assert.Equal("1", group.Values[0].Values![1].Id);
44     Assert.Equal("label", group.Values[1].Values![0].Id);
45     Assert.Equal("one", group.Values[1].Values![1].Id);
46 }
47
48 [Fact]
49 public void ParenthesesKeepSeveralRecordsSeparate()
50 {
51     AssertFormat(
52         "value (\n (id \"1\" label \"one\")\n (id \"2\" label \"two\")\n)",
53         "(value ((id 1 label one) (id 2 label two)))");
54 }
55
56 [Fact]
57 public void ParenthesesNestDeeply()
58 {
59     AssertFormat("outer (\n inner (\n      x 1\n      y 2\n  )\n z 3\n)", "(outer ((inner
↪ ((x 1) (y 2))) (z 3)))");
60 }
61
62 [Fact]
63 public void SingleLineParenthesesAreUnchanged()
64 {
65     AssertFormat("(a b c)", "(a b c)");
66     AssertFormat("(1: 2 3)", "(1: 2 3)");
67     AssertFormat("(a: b c)", "(a: b c)");
68     AssertFormat("((a b))", "((a b))");
69     AssertFormat("(a)", "(a)");
70     AssertFormat("()", "()");
71 }
72
73 [Fact]
74 public void ParenthesesWithIndentedIdSyntax()
75 {
76     AssertFormat("( \n  a: \n    b \n    c \n)", "(a: b c)");
77 }
78
79 [Fact]
80 public void EmployeeRecordsKeepTheirFields()
81 {
82     var source = "empInfo\n employees:\n      (\n          name (James Kirk)\n          age 40\n
↪ )\n"
83         + "      (\n          name (Jean-Luc Picard)\n          age 45\n      )";
84     var expected = "(empInfo)\n((empInfo) (employees: ((name (James Kirk)) (age 40)))"
85         + " ((name (Jean-Luc Picard)) (age 45)))");
86     AssertFormat(source, expected);
87 }
88 }
89 }

```

1.32 ./Link.Foundation.Links.Notation.Tests/NestedParserTests.cs

```

1  using System;
2  using System.Linq;
3  using Xunit;
4
5  namespace Link.Foundation.Links.Notation.Tests
6  {
7      public static class NestedParserTests
8      {
9          [Fact]
10         public static void SignificantWhitespaceTest()
11         {
12             var source = @"
13 users
14     user1
15         id
16             43
17         name
18             first
19                 John
20             last
21                 Williams
22         location
23             New York
24         age
25             23
26     user2

```

```

27         id
28         56
29         name
30         first
31         Igor
32         middle
33         Petrovich
34         last
35         Ivanov
36         location
37         Moscow
38         age
39         20";
40         var target = @"(users)
41 ((users) (user1))
42 (((users) (user1)) (id))
43 (((users) (user1)) (id)) (43))
44 (((users) (user1)) (name))
45 (((users) (user1)) (name)) (first))
46 (((users) (user1)) (name)) (first)) (John))
47 (((users) (user1)) (name)) (last))
48 (((users) (user1)) (name)) (last)) (Williams))
49 ((users) (user1)) (location))
50 (((users) (user1)) (location)) (New York))
51 ((users) (user1)) (age))
52 (((users) (user1)) (age)) (23))
53 ((users) (user2))
54 (((users) (user2)) (id))
55 (((users) (user2)) (id)) (56))
56 ((users) (user2)) (name))
57 (((users) (user2)) (name)) (first))
58 (((users) (user2)) (name)) (first)) (Igor))
59 (((users) (user2)) (name)) (middle))
60 (((users) (user2)) (name)) (middle)) (Petrovich))
61 (((users) (user2)) (name)) (last))
62 (((users) (user2)) (name)) (last)) (Ivanov))
63 ((users) (user2)) (location))
64 (((users) (user2)) (location)) (Moscow))
65 ((users) (user2)) (age))
66 (((users) (user2)) (age)) (20))";
67         var parser = new Parser();
68         var links = parser.Parse(source);
69         var formattedLinks = links.Format();
70         Assert.Equal(target, formattedLinks);
71     }
72
73     [Fact]
74     public static void SimpleSignificantWhitespaceTest()
75     {
76         var source = @"a
77 b
78 c";
79         var target = @"(a)
80 ((a) (b))
81 ((a) (c))";
82         var parser = new Parser();
83         var links = parser.Parse(source);
84         var formattedLinks = links.Format();
85         Assert.Equal(target, formattedLinks);
86     }
87
88     [Fact]
89     public static void TwoSpacesSizedWhitespaceTest()
90     {
91         var source = @"
92 users
93     user1";
94         var target = @"(users)
95 ((users) (user1))";
96         var parser = new Parser();
97         var links = parser.Parse(source);
98         var formattedLinks = links.Format();
99         Assert.Equal(target, formattedLinks);
100     }
101
102     [Fact]
103     public static void ParseNestedStructureWithIndentationTest()
104     {
105         var source = @"parent
106 child1

```

```

107     child2";
108         var parser = new Parser();
109         var links = parser.Parse(source);
110         Assert.NotNull(links);
111         // Should create 3 links: (parent), (parent child1), (parent child2)
112         Assert.Equal(3, links.Count);
113     }
114
115     [Fact]
116     public static void IndentationConsistencyTest()
117     {
118         // Test that indentation must be consistent
119         var input = @"parent
120     child1
121     child2"; // Inconsistent indentation
122         var result = new Parser().Parse(input);
123         // This should parse but child2 won't be a child of parent due to different
124     ↪ indentation
125         Assert.NotEmpty(result);
126     }
127
128     [Fact]
129     public static void IndentationBasedChildrenTest()
130     {
131         var input = @"parent
132     child1
133     child2
134     grandchild";
135         var parser = new Parser();
136         var result = parser.Parse(input);
137         Assert.NotNull(result);
138         // Expected flattened links:
139         // (parent), (parent child1), (parent child2), ((parent child2) grandchild)
140         Assert.Equal(4, result.Count);
141     }
142
143     [Fact]
144     public static void ComplexIndentationTest()
145     {
146         var input = @"root
147     level1a
148     level2a
149     level2b
150     level1b
151     level2c";
152         var parser = new Parser();
153         var result = parser.Parse(input);
154         Assert.NotNull(result);
155         // Expected flattened links:
156         // (root), (root level1a), ((root level1a) level2a), ((root level1a) level2b),
157         // (root level1b), ((root level1b) level2c)
158         Assert.Equal(6, result.Count);
159     }
160
161     [Fact]
162     public static void NestedLinksTest()
163     {
164         var input = "(1: (2: (3: 3)))";
165         var parser = new Parser();
166         var parsed = parser.Parse(input);
167         Assert.NotEmpty(parsed);
168
169         // Validate regular formatting
170         var output = parsed.Format();
171         Assert.NotEmpty(output);
172
173         // Validate that the structure is properly nested
174         Assert.Single(parsed);
175     }
176
177     [Fact]
178     public static void IndentationParserTest()
179     {
180         var input = "parent\n  child1\n  child2";
181         var parser = new Parser();
182         var result = parser.Parse(input);
183         Assert.NotEmpty(result);
184         // Should have parent link
185         var parentLink = result.FirstOrDefault(l => l.Id == "parent");

```

```

185     }
186
187     [Fact]
188     public static void NestedIndentationParserTest()
189     {
190         var input = "parent\n  child\n    grandchild";
191         var parser = new Parser();
192         var result = parser.Parse(input);
193         Assert.NotEmpty(result);
194         // Should create nested structure with proper hierarchy
195         Assert.True(result.Count >= 1);
196     }
197
198     [Fact]
199     public static void ThreeLevelNestingRoundtripTest()
200     {
201         var input = "(1: (2: (3: 3)))";
202         var parser = new Parser();
203         var parsed = parser.Parse(input);
204
205         // Validate round-trip
206         var output = parsed.Format();
207         Assert.Equal(input, output);
208     }
209
210     [Fact]
211     public static void DeepNestedStructureRoundtripTest()
212     {
213         var input = "(a: (b: (c: (d: d))))";
214         var parser = new Parser();
215         var parsed = parser.Parse(input);
216
217         // Validate round-trip
218         var output = parsed.Format();
219         Assert.Equal(input, output);
220     }
221
222     [Fact]
223     public static void MultipleNestedLinksRoundtripTest()
224     {
225         var input = "(parent: (child1: value1) (child2: value2))";
226         var parser = new Parser();
227         var parsed = parser.Parse(input);
228
229         // Validate round-trip
230         var output = parsed.Format();
231         Assert.Equal(input, output);
232     }
233 }
234 }

```

1.33 ./Link.Foundation.Links.Notation.Tests/NestedSelfReferenceTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class NestedSelfReferenceTests
7      {
8          [Fact]
9          public static void NestedSelfReferencedObjectInPairValueTest()
10         {
11             // Test case from PARSER BUG.md
12             // This should parse a dict with two pairs, where the second pair's value
13             // is itself a self-referenced dict definition (obj 1: dict ...)
14             var notation = "(obj 0: dict ((str bmFtZQ==) (str ZG1jdDE=)) ((str b3RoZXI=) (obj 1:
↪ dict ((str bmFtZQ==) (str ZG1jdDI=)) ((str b3RoZXI=) obj 0))))";
15
16             var parser = new Parser();
17             var links = parser.Parse(notation);
18
19             // Should parse exactly one top-level link
20             Assert.Single(links);
21
22             var link = links[0];
23
24             // Top-level link should have ID "obj 0"
25             Assert.Equal("obj 0", link.Id);
26

```

```

27 // Should have: type marker + 2 pairs = 3 values
28 Assert.Equal(3, link.Values!.Count);
29
30 // First value is the type marker "dict"
31 Assert.Equal("dict", link.Values[0].Id); -
32
33 // Second and third values are the two pairs
34 var pair1 = link.Values[1];
35 var pair2 = link.Values[2];
36
37 // Pair 1: ((str bmFtZQ==) (str ZGljdDE=))
38 // This is a parenthesized expression containing two sub-expressions
39 Assert.Null(pair1.Id);
40 Assert.Equal(2, pair1.Values!.Count);
41
42 // First element of pair1: (str bmFtZQ==)
43 Assert.Null(pair1.Values[0].Id);
44 Assert.Equal(2, pair1.Values![0].Values!.Count);
45 Assert.Equal("str", pair1.Values![0].Values![0].Id);
46 Assert.Equal("bmFtZQ==", pair1.Values![0].Values![1].Id); -
47
48 // Second element of pair1: (str ZGljdDE=)
49 Assert.Null(pair1.Values[1].Id);
50 Assert.Equal(2, pair1.Values![1].Values!.Count);
51 Assert.Equal("str", pair1.Values![1].Values![0].Id);
52 Assert.Equal("ZGljdDE=", pair1.Values![1].Values![1].Id);
53
54 // Pair 2: ((str b3RoZXI=) (obj 1: dict ...))
55 // This is a parenthesized expression containing two sub-expressions
56 Assert.Null(pair2.Id);
57 Assert.Equal(2, pair2.Values!.Count);
58
59 // First element of pair2: (str b3RoZXI=)
60 Assert.Null(pair2.Values[0].Id);
61 Assert.Equal(2, pair2.Values![0].Values!.Count);
62 Assert.Equal("str", pair2.Values![0].Values![0].Id);
63 Assert.Equal("b3RoZXI=", pair2.Values![0].Values![1].Id);
64
65 // Second element of pair2: (obj 1: dict ((str bmFtZQ==) (str ZGljdDI=)) ((str
↪ b3RoZXI=) obj 0))
66 // THIS IS THE KEY TEST - obj 1 should have its ID preserved
67 var obj1 = pair2.Values[1];
68 Assert.Equal("obj 1", obj1.Id);
69 Assert.NotNull(obj1.Values);
70 Assert.Equal(3, obj1.Values.Count); // type marker + 2 pairs
71
72 // obj 1's type marker
73 Assert.Equal("dict", obj1.Values[0].Id);
74
75 // obj 1's first pair: ((str bmFtZQ==) (str ZGljdDI=))
76 var obj1Pair1 = obj1.Values[1];
77 Assert.Equal(2, obj1Pair1.Values!.Count);
78 Assert.Null(obj1Pair1.Values[0].Id);
79 Assert.Equal(2, obj1Pair1.Values![0].Values!.Count);
80 Assert.Equal("str", obj1Pair1.Values![0].Values![0].Id);
81 Assert.Equal("bmFtZQ==", obj1Pair1.Values![0].Values![1].Id);
82 Assert.Null(obj1Pair1.Values[1].Id);
83 Assert.Equal(2, obj1Pair1.Values![1].Values!.Count);
84 Assert.Equal("str", obj1Pair1.Values![1].Values![0].Id);
85 Assert.Equal("ZGljdDI=", obj1Pair1.Values![1].Values![1].Id);
86
87 // obj 1's second pair: ((str b3RoZXI=) obj 0) - reference back to obj 0
88 var obj1Pair2 = obj1.Values[2];
89 Assert.Equal(2, obj1Pair2.Values!.Count);
90 Assert.Null(obj1Pair2.Values[0].Id);
91 Assert.Equal(2, obj1Pair2.Values![0].Values!.Count);
92 Assert.Equal("str", obj1Pair2.Values![0].Values![0].Id);
93 Assert.Equal("b3RoZXI=", obj1Pair2.Values![0].Values![1].Id);
94 Assert.Equal("obj 0", obj1Pair2.Values[1].Id);
95 Assert.Null(obj1Pair2.Values[1].Values); // Just a reference, no nested values
96 }
97
98 [Fact]
99 public static void SelfReferenceAsDirectChildWorksCorrectlyTest()
100 {
101 // This pattern should work (and did work before)
102 var notation = "(obj 0: list (int 1) (int 2) (obj 1: list (int 3) (int 4) obj 0))";
103
104 var parser = new Parser();

```

```

105     var links = parser.Parse(notation);
106
107     Assert.Single(links);
108     Assert.Equal("obj 0", links[0].Id);
109     Assert.Equal(4, links[0].Values!.Count); // list + 1 + 2 + obj 1
110
111     // The fourth value should be obj 1 with a self-reference
112     var obj1 = links[0].Values![3];
113     Assert.Equal("obj 1", obj1.Id);
114     Assert.Equal(4, obj1.Values!.Count); // list + 3 + 4 + obj 0
115     Assert.Equal("obj 0", obj1.Values[3].Id); // Reference to obj 0
116 }
117 }
118 }

```

1.34 ./Link.Foundation.Links.Notation.Tests/NestingDepthTests.cs

```

1  using System;
2  using System.Linq;
3  using System.Threading;
4  using Xunit;
5
6  namespace Link.Foundation.Links.Notation.Tests
7  {
8      /// <summary>
9      /// Reading nested groups must take time that grows with the size of the document,
10     /// not with two to the power of its nesting depth
11     /// (<see href="https://github.com/link-foundation/links-notation/issues/314">#314</see>).
12     /// Before the fix, every level of nesting doubled the work: a document nested six
13     /// levels deep took seconds. Each case runs on its own thread and fails when it has
14     /// not finished within the budget, so a regression fails instead of hanging.
15     /// </summary>
16     public class NestingDepthTests
17     {
18         /// Generous enough for a debug build on a slow machine, and far below what the
19         /// exponential reading took at the depths used here.
20         private static readonly TimeSpan Budget = TimeSpan.FromSeconds(5);
21
22         /// The depth used for the shapes that used to take exponential time.
23         private const int Deep = 32;
24
25         /// A depth that shows the reading stays linear well past the shallow cases.
26         private const int VeryDeep = 256;
27
28         private static string Closed(int depth) => new string('(', depth) + "a" + new
↪ string(')', depth);
29
30         private static string ValueAfter(int depth) => new string('(', depth) + "a" +
↪ string.Concat(Enumerable.Repeat(" b", depth));
31
32         private static string Unclosed(int depth) => new string('(', depth) + "a";
33
34         /// Unclosed groups on lines that are each indented one space deeper.
35         private static string IndentedUnclosed(int depth) =>
36         ↪ string.Join("\n", Enumerable.Range(0, depth).Select(level => new string(' ', level)
↪ + "(a)"));
37
38         /// Formats the parsed document, or names the exception parsing raised.
39         private static string ReadWithinBudget(string what, string source)
40         {
41             string? outcome = null;
42             var reader = new Thread(() =>
43             {
44                 try
45                 {
46                     outcome = new Parser().Parse(source).Format();
47                 }
48                 catch (Exception error)
49                 {
50                     outcome = error.GetType().Name;
51                 }
52             }, 512 * 1024 * 1024)
53             {
54                 // A reader that runs past the budget must not keep the test run alive.
55                 IsBackground = true,
56             };
57             reader.Start();
58             Assert.True(reader.Join(Budget), $"{what} did not finish within {Budget}");
59             return outcome!;
60         }
61     }
62 }

```

```

61
62     [Fact]
63     public void ClosedGroupsReadInLinearTime()
64     {
65         var source = Closed(Deep);
66         Assert.Equal(source, ReadWithinBudget("closed groups", source));
67     }
68
69     [Fact]
70     public void ValuesAfterGroupsReadInLinearTime()
71     {
72         var source = ValueAfter(Deep);
73         Assert.Equal($"({source})", ReadWithinBudget("values after groups", source));
74     }
75
76     [Fact]
77     public void UnclosedGroupsFailInLinearTime()
78     {
79         Assert.Equal(nameof(ParseException), ReadWithinBudget("unclosed groups",
↪ Unclosed(Deep)));
80     }
81
82     [Fact]
83     public void UnclosedGroupsOnIndentedLinesFailInLinearTime()
84     {
85         Assert.Equal(nameof(ParseException), ReadWithinBudget("unclosed indented groups",
↪ IndentedUnclosed(Deep)));
86     }
87
88     [Fact]
89     public void ValuesAfterVeryDeepGroupsReadInLinearTime()
90     {
91         var source = ValueAfter(VeryDeep);
92         Assert.Equal($"({source})", ReadWithinBudget("very deep values after groups",
↪ source));
93     }
94
95     [Fact]
96     public void GroupFollowedByValuesKeepsItsStructure()
97     {
98         var cases = new[]
99         {
100             new[] { "(a b", "((a b)" },
101             new[] { "(a (b) c", "((a (b) c)" },
102             new[] { "((a) b) c", "(((a) b) c)" },
103             new[] { "(a: b) c", "((a: b) c)" },
104             new[] { "(a)\n(b) c", "(a)\n((b) c)" },
105             new[] { "x\n (a) b\n (c)", "(x)\n((x) ((a) b))\n((x) (c))" },
106         };
107
108         var parser = new Parser();
109         foreach (var testCase in cases)
110         {
111             Assert.Equal(testCase[1], parser.Parse(testCase[0]).Format());
112         }
113     }
114 }
115 }

```

1.35 ./Link.Foundation.Links.Notation.Tests/ParseErrorPositionTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      /// <summary>
7      /// A parse error has to say where the document stopped making sense.
8      /// </summary>
9      /// <remarks>
10     /// The positions asserted here are the ones the JavaScript and Rust ports report for
11     /// the same input, so the implementations can be held to the same contract
12     /// (https://github.com/link-foundation/links-notation/issues/302).
13     /// </remarks>
14     public static class ParseErrorPositionTests
15     {
16         private static ParseException SyntaxError(string document) =>
17             Assert.Throws<ParseException>(() => new Parser().Parse(document));
18     }

```

```

19 [Fact]
20 public static void ReportsTheLineAndColumnOfTheDefectTest()
21 {
22     // The example from the issue: the defect is the second colon on line 2,
23     // and the two lines after it are fine.
24     var error = SyntaxError("ci gate x\nstage: rust: nexttest\nnext stage\n clippy");
25
26     Assert.Equal(2, error.Line);
27     Assert.Equal(12, error.Column);
28     Assert.Equal(':', error.Found);
29 }
30
31 [Fact]
32 public static void OffsetAgreesWithTheOtherImplementationsTest()
33 {
34     // JavaScript and Rust report offset 21, line 2, column 12 for this document.
35     var error = SyntaxError("ci gate x\nstage: rust: nexttest\n");
36
37     Assert.Equal(21, error.Offset);
38     Assert.Equal(2, error.Line);
39     Assert.Equal(12, error.Column);
40 }
41
42 [Fact]
43 public static void ReportsTheLineALateDefectIsOnTest()
44 {
45     var error = SyntaxError("a\nb\nc\nd\ne: f: g\nh\n");
46
47     Assert.Equal(5, error.Line);
48     Assert.Equal(5, error.Column);
49     Assert.Equal("e: f: g", error.LineText);
50 }
51
52 [Fact]
53 public static void ReportsTheEndOfTheDocumentWhenAGroupIsNeverClosedTest()
54 {
55     var error = SyntaxError("a (b\n");
56
57     Assert.Equal(5, error.Offset);
58     Assert.Equal(2, error.Line);
59     Assert.Equal(1, error.Column);
60     Assert.Null(error.Found);
61     Assert.Contains("end of input", error.Message);
62 }
63
64 [Fact]
65 public static void ReportsAnUnmatchedClosingParenthesisTest()
66 {
67     var error = SyntaxError("a b)\n");
68
69     Assert.Equal(3, error.Offset);
70     Assert.Equal(1, error.Line);
71     Assert.Equal(4, error.Column);
72     Assert.Equal(')', error.Found);
73 }
74
75 [Fact]
76 public static void MessagePointsACaretAtTheOffendingCharacterTest()
77 {
78     var error = SyntaxError("ci gate x\nstage: rust: nexttest\n");
79
80     Assert.Equal("line 2, column 12: unexpected \":\"", error.Summary);
81     Assert.Equal("2 | stage: rust: nexttest\n |", error.Snippet);
82     Assert.Equal(
83         "Syntax error at line 2, column 12: unexpected \":\"\n2 | stage: rust: nexttest\n",
84         error.Message);
85 }
86
87 [Fact]
88 public static void MessageQuotesOneLineRatherThanTheRestOfTheDocumentTest()
89 {
90     var document = "ci gate x\nstage: rust: nexttest\n"
91         + string.Concat(System.Linq.Enumerable.Repeat("trailing line\n", 500));
92
93     var error = SyntaxError(document);
94
95     Assert.Contains("line 2, column 12", error.Message);
96     Assert.DoesNotContain("trailing line", error.Message);

```

```

97         Assert.True(error.Message.Length < 200, $"message is {error.Message.Length}
↪ characters");
98     }
99
100     [Fact]
101     public static void MessageOfALongLineStaysAMessageTest()
102     {
103         var document = new string('a', 400) + ": " + new string('b', 400) + ": c";
104
105         var error = SyntaxError(document);
106
107         Assert.Equal(1, error.Line);
108         Assert.Equal(803, error.Column);
109         Assert.Contains("...", error.Message);
110         Assert.True(error.Message.Length < 300, $"message is {error.Message.Length}
↪ characters");
111     }
112
113     [Fact]
114     public static void KeepsCatchingCodeThatExpectsAFormatExceptionWorkingTest()
115     {
116         // The generated parser used to raise a bare FormatException; code that catches
117         // it keeps working, and now reads a position off the exception it catches.
118         var error = Assert.Throws<ParseException>(() => new Parser().Parse("a: b: c"));
119
120         Assert.IsAssignableFrom<FormatException>(error);
121         Assert.Equal(4, error.Offset);
122     }
123
124     [Fact]
125     public static void DoesNotMentionTheGrammarInternalsTest()
126     {
127         var error = SyntaxError("a: b: c");
128
129         Assert.DoesNotContain("Failed to parse", error.Message);
130         Assert.DoesNotContain("document", error.Message);
131     }
132
133     [Fact]
134     public static void ADocumentThatParsesReportsNothingTest()
135     {
136         var links = new Parser().Parse("a: b\n c: d\n");
137
138         Assert.Equal(2, links.Count);
139     }
140 }
141 }

```

1.36 ./Link.Foundation.Links.Notation.Tests/SingleLineParserTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class SingleLineParserTests
7      {
8          [Fact]
9          public static void SingleLinkTest()
10         {
11             var source = @"(address: source target)";
12             var parser = new Parser();
13             var links = parser.Parse(source);
14             var target = links.Format();
15             Assert.Equal(source, target);
16         }
17
18         [Fact]
19         public static void TripletSingleLinkTest()
20         {
21             var source = @"(papa has car)";
22             var parser = new Parser();
23             var links = parser.Parse(source);
24             var target = links.Format();
25             Assert.Equal(source, target);
26         }
27
28         [Fact]
29         public static void BugTest1()
30         {

```

```

31     var source = @"(ignore conan-center-index repository)";
32     var links = new Parser().Parse(source);
33     var target = links.Format();
34     Assert.Equal(source, target);
35 }
36
37 [Fact]
38 public static void QuotedReferencesTest()
39 {
40     var source = @"(a: 'b' "c")";
41     var target = @"(a: b c)";
42     var parser = new Parser();
43     var links = parser.Parse(source);
44     var formattedLinks = links.Format();
45     Assert.Equal(target, formattedLinks);
46 }
47
48 [Fact]
49 public static void QuotedReferencesWithSpacesTest()
50 {
51     var source = @"('a a': 'b b' "c c")";
52     var target = @"('a a': 'b b' 'c c')";
53     var parser = new Parser();
54     var links = parser.Parse(source);
55     var formattedLinks = links.Format();
56     Assert.Equal(target, formattedLinks);
57 }
58
59 [Fact]
60 public static void ParseSimpleReferenceTest()
61 {
62     var source = "test";
63     var parser = new Parser();
64     var links = parser.Parse(source);
65     Assert.NotNull(links);
66     Assert.Single(links);
67     // Simple reference creates a singlet link with null Id and one value
68     Assert.Null(links[0].Id);
69     Assert.NotNull(links[0].Values);
70     Assert.Single(links[0].Values!);
71     Assert.Equal("test", links[0].Values?[0].Id);
72     Assert.Null(links[0].Values?[0].Values);
73 }
74
75 [Fact]
76 public static void ParseReferenceWithColonAndValuesTest()
77 {
78     var source = "parent: child1 child2";
79     var parser = new Parser();
80     var links = parser.Parse(source);
81     Assert.NotNull(links);
82     Assert.Single(links);
83     Assert.Equal("parent", links[0].Id);
84     Assert.NotNull(links[0].Values);
85     Assert.Equal(2, links[0].Values!.Count);
86     Assert.Equal("child1", links[0].Values![0].Id);
87     Assert.Equal("child2", links[0].Values![1].Id);
88 }
89
90 [Fact]
91 public static void ParseMultilineLinkTest()
92 {
93     var source = "(parent: child1 child2)";
94     var parser = new Parser();
95     var links = parser.Parse(source);
96     Assert.NotNull(links);
97     Assert.Single(links);
98     Assert.Equal("parent", links[0].Id);
99     Assert.NotNull(links[0].Values);
100    Assert.Equal(2, links[0].Values!.Count);
101 }
102
103 [Fact]
104 public static void ParseValuesOnlyTest()
105 {
106     var source = ": value1 value2";
107     var parser = new Parser();
108     // Standalone ':' is now forbidden and should throw an exception

```

```

109     Assert.Throws<ParseException>(() => parser.Parse(source));
110 }
111
112 [Fact]
113 public static void SingletLinkTest()
114 {
115     // Test singlet link
116     var input = "(singlet)";
117     var result = new Parser().Parse(input);
118     Assert.Single(result);
119     Assert.Null(result[0].Id);
120     Assert.NotNull(result[0].Values);
121     Assert.Single(result[0].Values!);
122     Assert.Equal("singlet", result[0].Values?[0].Id);
123     Assert.Null(result[0].Values?[0].Values);
124 }
125
126 [Fact]
127 public static void ValueLinkTest()
128 {
129     // Test value link
130     var input = "(value1 value2 value3)";
131     var result = new Parser().Parse(input);
132     Assert.NotEmpty(result);
133 }
134
135 [Fact]
136 public static void QuotedReferencesWithSpecialCharsTest()
137 {
138     // Test quoted references
139     var input = @"("id with spaces": "value with spaces")";
140     var result = new Parser().Parse(input);
141     Assert.NotEmpty(result);
142 }
143
144 [Fact]
145 public static void SingleQuotedReferencesTest()
146 {
147     // Test single-quoted references
148     var input = "('id': 'value')";
149     var result = new Parser().Parse(input);
150     Assert.NotEmpty(result);
151 }
152
153 [Fact]
154 public static void ParseQuotedReferencesValuesOnlyTest()
155 {
156     var source = "\"has space\" 'has:colon'";
157     var parser = new Parser();
158     var links = parser.Parse(source);
159     Assert.NotNull(links);
160     Assert.Single(links);
161     Assert.Null(links[0].Id);
162     Assert.NotNull(links[0].Values);
163     Assert.Equal(2, links[0].Values!.Count);
164     Assert.Equal("has space", links[0].Values![0].Id);
165     Assert.Equal("has:colon", links[0].Values![1].Id);
166     var formatted = links.Format();
167     Assert.Equal("('has space' 'has:colon')", formatted);
168 }
169
170 [Fact]
171 public static void NestedLinksTest()
172 {
173     // Test nested links
174     var input = "(outer: (inner: value))";
175     var result = new Parser().Parse(input);
176     Assert.NotEmpty(result);
177 }
178
179 [Fact]
180 public static void HyphenatedIdentifiersTest()
181 {
182     // Test support for hyphenated identifiers like in BugTest1
183     var source = @"(conan-center-index: repository info)";
184     var parser = new Parser();
185     var links = parser.Parse(source);
186     var target = links.Format();

```

```

187     Assert.Equal(source, target);
188 }
189
190 [Fact]
191 public static void MultipleWordsInQuotesTest()
192 {
193     var source = @"("New York": city state)";
194     var parser = new Parser();
195     var links = parser.Parse(source);
196     var target = links.Format();
197     // Should preserve quotes for multi-word references
198     Assert.Contains("New York", target);
199 }
200
201 [Fact]
202 public static void SpecialCharactersInQuotesTest()
203 {
204     var input = @"("key:with:colons": "value(with)parense")";
205     var result = new Parser().Parse(input);
206     Assert.NotEmpty(result);
207
208     input = @"('key with spaces': 'value: with special chars')";
209     result = new Parser().Parse(input);
210     Assert.NotEmpty(result);
211 }
212
213 [Fact]
214 public static void DeeplyNestedTest()
215 {
216     var input = "(a: (b: (c: (d: (e: value))))))";
217     var result = new Parser().Parse(input);
218     Assert.NotEmpty(result);
219 }
220
221 [Fact]
222 public static void SingleLineWithIdTest()
223 {
224     // Test single-line link with id
225     var input = "id: value1 value2";
226     var result = new Parser().Parse(input);
227     Assert.NotEmpty(result);
228 }
229
230 [Fact]
231 public static void SingleLineWithoutIdTest()
232 {
233     // Test link without id (single-line) - now forbidden
234     var input = ": value1 value2";
235     Assert.Throws<ParseException>(() => new Parser().Parse(input));
236 }
237
238 [Fact]
239 public static void MultilineWithoutIdTest()
240 {
241     // Test link without id (multi-line) - now forbidden
242     var input = "(: value1 value2)";
243     Assert.Throws<ParseException>(() => new Parser().Parse(input));
244 }
245
246 [Fact]
247 public static void SimpleRefTest()
248 {
249     var input = "simple ref";
250     var result = new Parser().Parse(input);
251     Assert.NotEmpty(result);
252 }
253
254 [Fact]
255 public static void MultiLineLinkWithIdTest()
256 {
257     var input = "(id: value1 value2)";
258     var result = new Parser().Parse(input);
259     Assert.NotEmpty(result);
260 }
261
262 [Fact]
263 public static void LinkWithoutIdMultiLineTest()
264 {
265     var input = "(: value1 value2)";

```

```

266     Assert.Throws<ParseException>(() => new Parser().Parse(input));
267 }
268
269 [Fact]
270 public static void SimpleReferenceParserTest()
271 {
272     var input = "hello";
273     var result = new Parser().Parse(input);
274     Assert.Single(result);
275     Assert.Null(result[0].Id);
276     Assert.NotNull(result[0].Values);
277     Assert.Single(result[0].Values!);
278     Assert.Equal("hello", result[0].Values?[0].Id);
279 }
280
281 [Fact]
282 public static void QuotedReferenceParserTest()
283 {
284     var input = "\"hello world\"";
285     var result = new Parser().Parse(input);
286     Assert.Single(result);
287     Assert.Null(result[0].Id);
288     Assert.NotNull(result[0].Values);
289     Assert.Single(result[0].Values!);
290     Assert.Equal("hello world", result[0].Values?[0].Id);
291 }
292
293 [Fact]
294 public static void SingletLinkParserTest()
295 {
296     var input = "(singlet)";
297     var result = new Parser().Parse(input);
298     Assert.Single(result);
299     Assert.Null(result[0].Id);
300     Assert.NotNull(result[0].Values);
301     Assert.Single(result[0].Values!);
302     Assert.Equal("singlet", result[0].Values?[0].Id);
303     Assert.Null(result[0].Values?[0].Values);
304 }
305
306 [Fact]
307 public static void ValueLinkParserTest()
308 {
309     var input = "(a b c)";
310     var result = new Parser().Parse(input);
311     Assert.Single(result);
312     Assert.Null(result[0].Id);
313     Assert.Equal(3, result[0].Values?.Count);
314 }
315
316 [Fact]
317 public static void ParseQuotedReferencesTest()
318 {
319     var input = "(\"quoted id\": \"value with spaces\")";
320     var parser = new Parser();
321     var result = parser.Parse(input);
322
323     Assert.NotEmpty(result);
324     var formatted = result.Format();
325     Assert.Contains("quoted id", formatted);
326     Assert.Contains("value with spaces", formatted);
327 }
328
329 [Fact]
330 public static void LinkWithoutIdSingleLineTest()
331 {
332     // Test that standalone ':' is forbidden and should throw
333     var input = ": value1 value2";
334     var parser = new Parser();
335
336     // C# parser forbids this syntax (like JS/Rust)
337     Assert.Throws<ParseException>(() => parser.Parse(input));
338 }
339
340 [Fact]
341 public static void SingleLineLinkWithIdTest()
342 {
343     var input = "id: value1 value2";
344     var parser = new Parser();

```

```

345     var result = parser.Parse(input);
346
347     Assert.NotEmpty(result);
348     Assert.NotNull(result[0].Id);
349     Assert.Equal("id", result[0].Id);
350 }
351
352 [Fact]
353 public static void LinkWithIdTest()
354 {
355     var input = "(id: a b c)";
356     var parser = new Parser();
357     var result = parser.Parse(input);
358
359     Assert.Single(result);
360     Assert.Equal("id", result[0].Id);
361     Assert.Equal(3, result[0].Values?.Count);
362 }
363
364 [Fact]
365 public static void QuotedReferenceTest()
366 {
367     var input = "\"quoted value\"";
368     var parser = new Parser();
369     var result = parser.Parse(input);
370
371     Assert.NotEmpty(result);
372     Assert.NotNull(result[0].Values);
373     Assert.Single(result[0].Values!);
374     Assert.Equal("quoted value", result[0].Values![0].Id);
375 }
376
377 [Fact]
378 public static void SimpleReferenceTest()
379 {
380     var input = "simplereference";
381     var parser = new Parser();
382     var result = parser.Parse(input);
383
384     Assert.NotEmpty(result);
385 }
386
387 [Fact]
388 public static void SingleLineLinkTest()
389 {
390     var input = "myid: value1 value2";
391     var parser = new Parser();
392     var result = parser.Parse(input);
393
394     Assert.Single(result);
395     Assert.Equal("myid", result[0].Id);
396     Assert.Equal(2, result[0].Values?.Count);
397 }
398
399 [Fact]
400 public static void ValuesOnlyInParenthesesTest()
401 {
402     var input = "(value1 value2)";
403     var parser = new Parser();
404     var result = parser.Parse(input);
405
406     Assert.Single(result);
407     Assert.Null(result[0].Id);
408     Assert.Equal(2, result[0].Values?.Count);
409 }
410
411 [Fact]
412 public static void ParseValuesOnlyStandaloneColonTest()
413 {
414     // Test that standalone ':' is forbidden and should throw
415     var input = ": value1 value2";
416     var parser = new Parser();
417
418     // C# parser forbids this syntax (like JS/Rust)
419     Assert.Throws<ParseException>(() => parser.Parse(input));
420 }
421
422 [Fact]
423 public static void QuotedReferencesWithSpacesInLinkTest()

```

```

424     {
425         var input = "(id: \"value with spaces\")";
426         var parser = new Parser();
427         var result = parser.Parse(input);
428
429         Assert.Single(result);
430         Assert.Equal("id", result[0].Id);
431         Assert.Single(result[0].Values!);
432         Assert.Equal("value with spaces", result[0].Values?[0].Id);
433     }
434 }
435 }

```

1.37 ./Link.Foundation.Links.Notation.Tests/StreamParserTests.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Threading.Tasks;
5  using Xunit;
6
7  namespace Link.Foundation.Links.Notation.Tests
8  {
9      public static class StreamParserTests
10     {
11         private const string Document =
12             "first loves data\n" +
13             "# streamed comment\n" +
14             "profile:\n" +
15             "    name Ada\n" +
16             "    note \"line one\nline two\"\n" +
17             "(nested:\n    child value)\n" +
18             "last sees first";
19
20         private static string[] Render(IEnumerable<Link<string>> links) =>
21             links.Select(link => link.ToString()).ToArray();
22
23         [Fact]
24         public static void MatchesCanonicalParserOneSymbolAtATime()
25         {
26             var stream = new StreamParser();
27             foreach (var character in Document)
28             {
29                 stream.Write(character.ToString());
30             }
31
32             Assert.Equal(Render(new Parser().Parse(Document)), Render(stream.Finish()));
33         }
34
35         [Fact]
36         public static void EmitsOnlyCompleteRecords()
37         {
38             var seen = new List<Link<string>>();
39             var stream = new StreamParser();
40             stream.LinkParsed += seen.Add;
41
42             Assert.Empty(stream.Write("profile:\n    name Ada\n    note \"first\nsecond\"\n"));
43             var emitted = stream.Write("\n");
44
45             Assert.Single(emitted);
46             Assert.Equal(emitted, seen);
47         }
48
49         [Fact]
50         public static void SupportsLineChunksFinalRecordAndPosition()
51         {
52             var stream = new StreamParser();
53             foreach (var line in Document.SplitAfter('\n'))
54             {
55                 stream.Write(line);
56             }
57
58             Assert.Equal(Render(new Parser().Parse(Document)), Render(stream.Finish()));
59             Assert.Equal(Document.Length, stream.Position.Offset);
60             Assert.Equal(0, stream.Position.Buffered);
61         }
62
63         [Fact]
64         public static void SupportsDrainResetAndBoundedMemory()
65         {

```

```

66     var seen = new List<Link<string>>();
67     var stream = new StreamParser { Collect = false, MaxBufferSize = 8 };
68     stream.LinkParsed += seen.Add;
69     for (var index = 0; index < 100; index++) stream.Write("a\n");
70
71     Assert.Empty(stream.Drain());
72     Assert.Single(stream.Finish());
73     Assert.Equal(100, seen.Count);
74
75     stream.Reset();
76     Assert.Throws<ArgumentException>(() => { stream.Write("123456789"); });
77 }
78
79 [Fact]
80 public static void ProvidesLazyAdapters()
81 {
82     var actual = StreamParser.ParseChunks(new[] { "one link\n", "two link" });
83
84     Assert.Equal(new[] { "(one link)", "(two link)" }, Render(actual));
85 }
86
87 [Fact]
88 public static void RejectsWritesAfterFinish()
89 {
90     var stream = new StreamParser();
91     stream.Finish("one");
92
93     Assert.Throws<InvalidOperationException>(() => stream.Write("two"));
94 }
95
96 [Fact]
97 public static async Task ProvidesAnAsyncEnumerableAdapter()
98 {
99     static async IEnumerable<string> Chunks()
100     {
101         yield return "one link\n";
102         await Task.Yield();
103         yield return "two link";
104     }
105
106     var actual = new List<Link<string>>();
107     await foreach (var link in StreamParser.ParseChunksAsync(
108         Chunks(),
109         cancellationTokens: TestContext.Current.CancellationToken)) actual.Add(link);
110
111     Assert.Equal(new[] { "(one link)", "(two link)" }, Render(actual));
112 }
113 }
114
115 internal static class StreamingTestStringExtensions
116 {
117     public static IEnumerable<string> SplitAfter(this string value, char delimiter)
118     {
119         var start = 0;
120         for (var index = 0; index < value.Length; index++)
121         {
122             if (value[index] != delimiter) continue;
123             yield return value.Substring(start, index - start + 1);
124             start = index + 1;
125         }
126         if (start < value.Length) yield return value.Substring(start);
127     }
128 }
129

```

1.38 ./Link.Foundation.Links.Notation.Tests/TupleTests.cs

```

1  using System.Collections.Generic;
2  using Xunit;
3  using LinkType = Link.Foundation.Links.Notation.Link<string>;
4  using id = Link.Foundation.Links.Notation.id<string>;
5  using = Link.Foundation.Links.Notation. <string>;
6
7  namespace Link.Foundation.Links.Notation.Tests
8  {
9      public class TupleTests
10     {
11         [Fact]
12         public void TupleToLinkTest()
13         {
14             var source = @"(papa (lovesMama: loves mama))

```

```

15 (son lovesMama)
16 (daughter lovesMama)
17 (all (love mama));
18     var parser = new Parser();
19     var links = parser.Parse(source);
20     var targetFromString = links.Format();
21
22     IList<LinkType> constructedLinks = new List<LinkType>()
23     {
24         ("papa", ( )("lovesMama", "loves", "mama")),
25         ("son", "lovesMama"),
26         ("daughter", "lovesMama"),
27         ("all", ("love", "mama")),
28     };
29     var targetFromTuples = constructedLinks.Format();
30     Assert.Equal(targetFromString, targetFromTuples);
31 }
32
33 [Fact]
34 public void NamedTupleToLinkTest()
35 {
36     var source = @"(papa (lovesMama: loves mama))
37 (son lovesMama)
38 (daughter lovesMama)
39 (all (love mama));
40     var parser = new Parser();
41     var links = parser.Parse(source);
42     var targetFromString = links.Format();
43
44     IList<LinkType> constructedLinks = new List<LinkType>()
45     {
46         ("papa", ((id)"lovesMama", "loves", "mama")),
47         ("son", "lovesMama"),
48         ("daughter", "lovesMama"),
49         ("all", ("love", "mama")),
50     };
51     var targetFromTuples = constructedLinks.Format();
52     Assert.Equal(targetFromString, targetFromTuples);
53 }
54 }
55 }

```

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